DoMS
QUIET EXCELLENCE DEMONSTRATED

MS3520 : Operations Management I

Forecasting

Forecasting

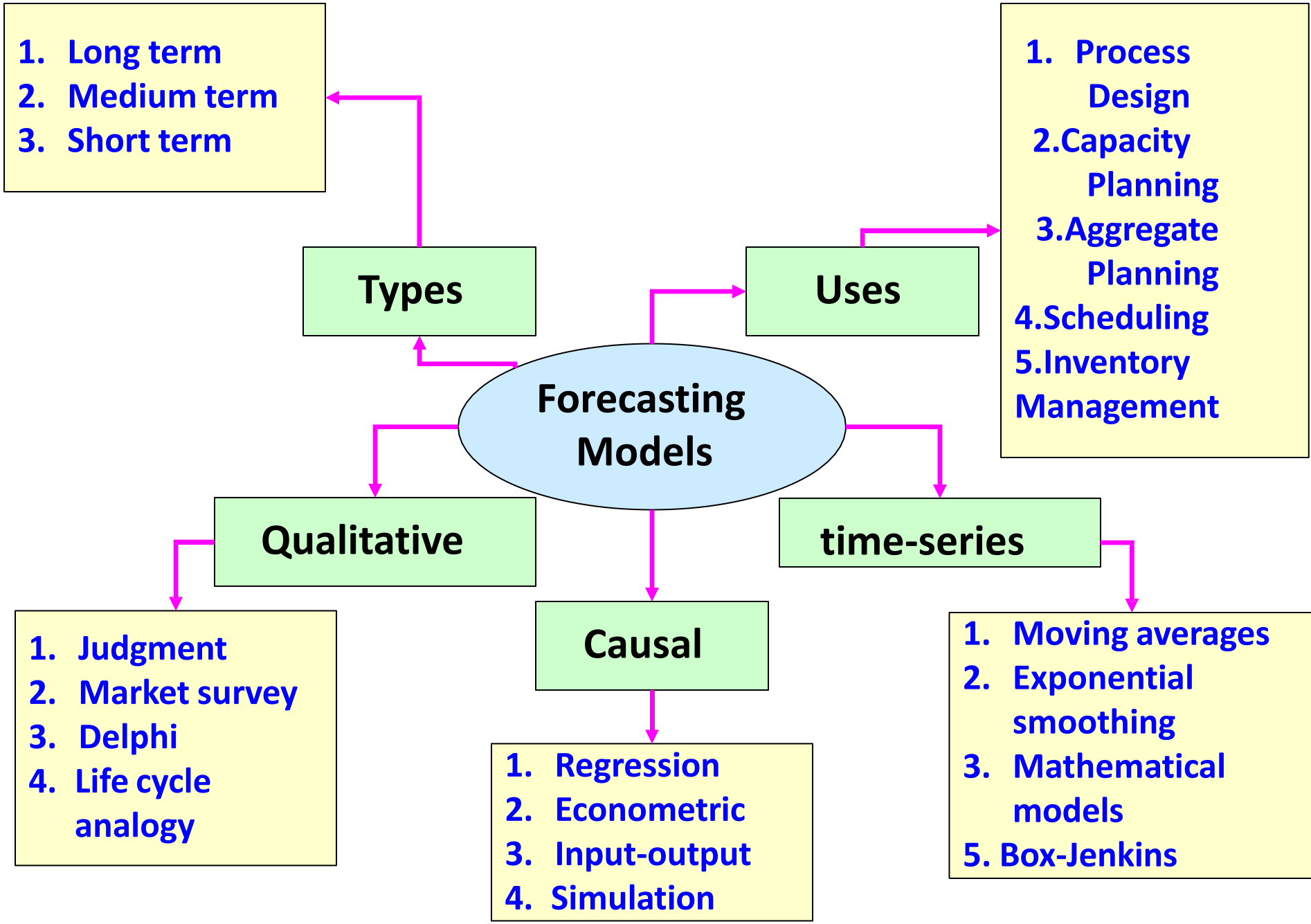
- Forecast:
  - An educated guess for the future based on objective analysis to the extent possible.
- Despite the presence of inaccuracies, some forecast is better than no forecast.
- Analytical forecast is better than intuitive forecast.
- Types of Forecast:
  - Demand - Common
  - Environmental - Social, Political, Economic
- Sales characteristics: RCTS
- Users:
  - Long range – Facilities
  - Medium range – Production Planning
  - Short range – Product Forecast

Principles of Forecasting

- Forecasts are rarely perfect.
  - Any prediction involves some uncertainty.
  - Goal of forecasting is to keep the forecast error as low as possible.
- Forecasts are more accurate for groups of items than for individual items.
  - Forecasting errors among items in a group usually have a cancelling effect.
- Forecasts are more accurate for shorter time horizons than longer ones.
  - Short-range forecasts must contend with fewer uncertainties than longer-range forecasts.

Steps in the Forecasting Process

- Determine the purpose of the forecast
- Establish a time horizon
- Evaluate and analyse appropriate data
- Select and test the forecasting model
- Generate the forecast
- Monitor the forecast errors


Forecasting Models
Types
1. Long term
2. Medium term
3. Short term
Uses
1. Process Design
2.Capacity Planning
3.Aggregate Planning
4.Scheduling
5.Inventory Management
Qualitative
1. Judgment
2. Market survey
3. Delphi
4. Life cycle analogy
Causal
1. Regression
2. Econometric
3. Input-output
4. Simulation
time-series
1. Moving averages
2. Exponential smoothing
3. Mathematical models
5. Box-Jenkins

Forecast Types Based on Source

- Subjective Opinion
- Indices of Business Activity
- Averages
- Statistical
- Combination
- Market Survey

Subjective Opinion

- Collective opinion of sales department
  - Advantage
    - People directly concerned
  - Disadvantage
    - Too much influence of immediate past
    - "Big Boss" opinion
    - Cannot blame anyone for goofs

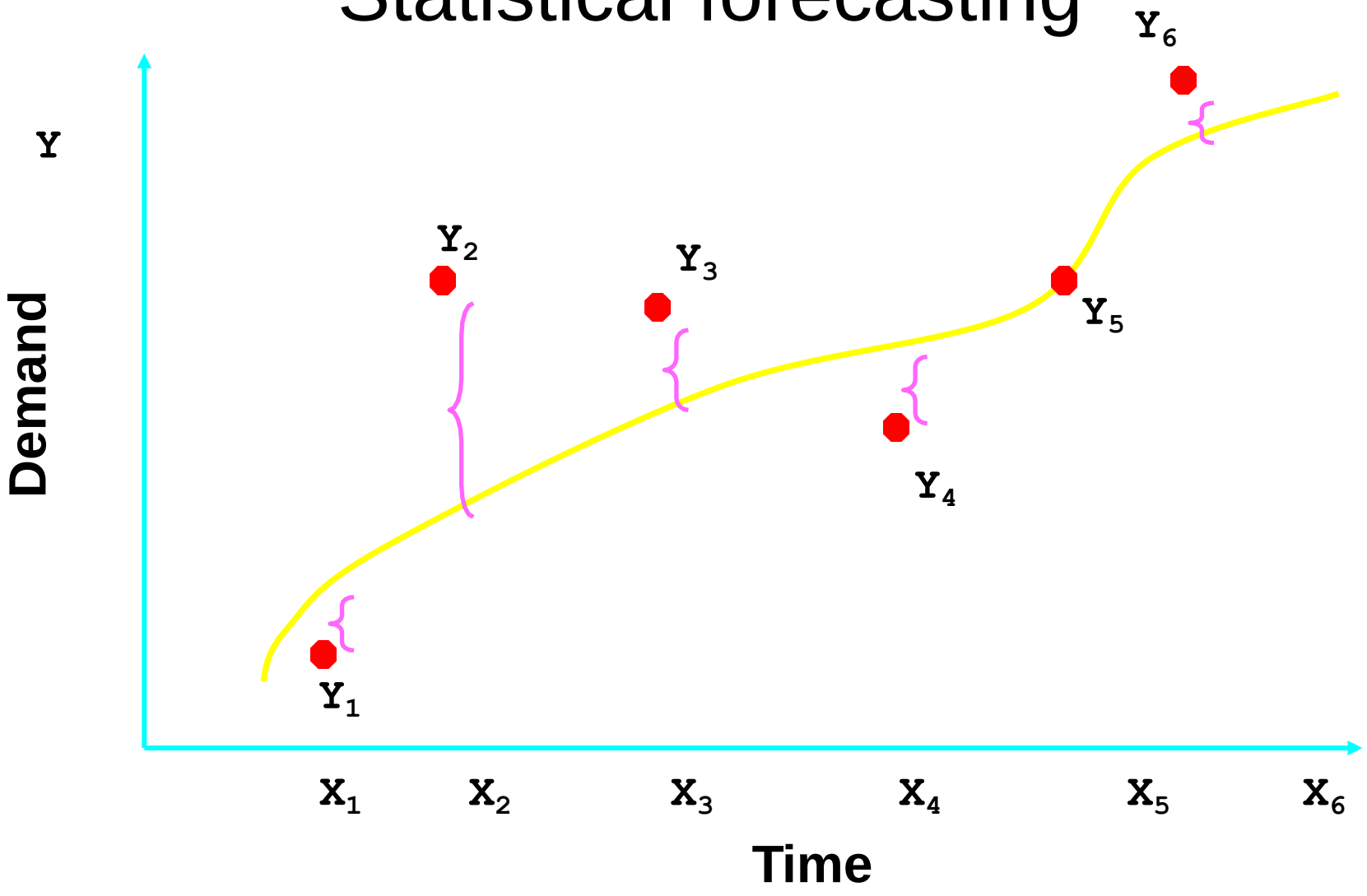
Indices of Business Activity

- Trend of gross national income, per capita income
- Index of construction etc.,
- Not very accurate

Averages

- Average of past sales
- Moving average
- Weighted average

Statistical forecasting



Method of least squares

- Let $\mathbf{X}_i = [X_{i1} \quad X_{i2} \quad X_{i3} \cdots X_{ik}]$ be a vector of values of k predictor variables for the ith observation among n observations.

- Let $\hat{Y}_i = f(\mathbf{X}_i)$ be the predicted response and Y_i be the actual response for the ith observation, respectively.

- Using the sum of deviations, i.e., $\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)$ could lead to incorrect inferences about the accuracy of forecasts due to the effect of canceling out.

- So we take the square of the deviations, find their sum, i.e., $\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2$ and minimize it to determine the predictor coefficients.

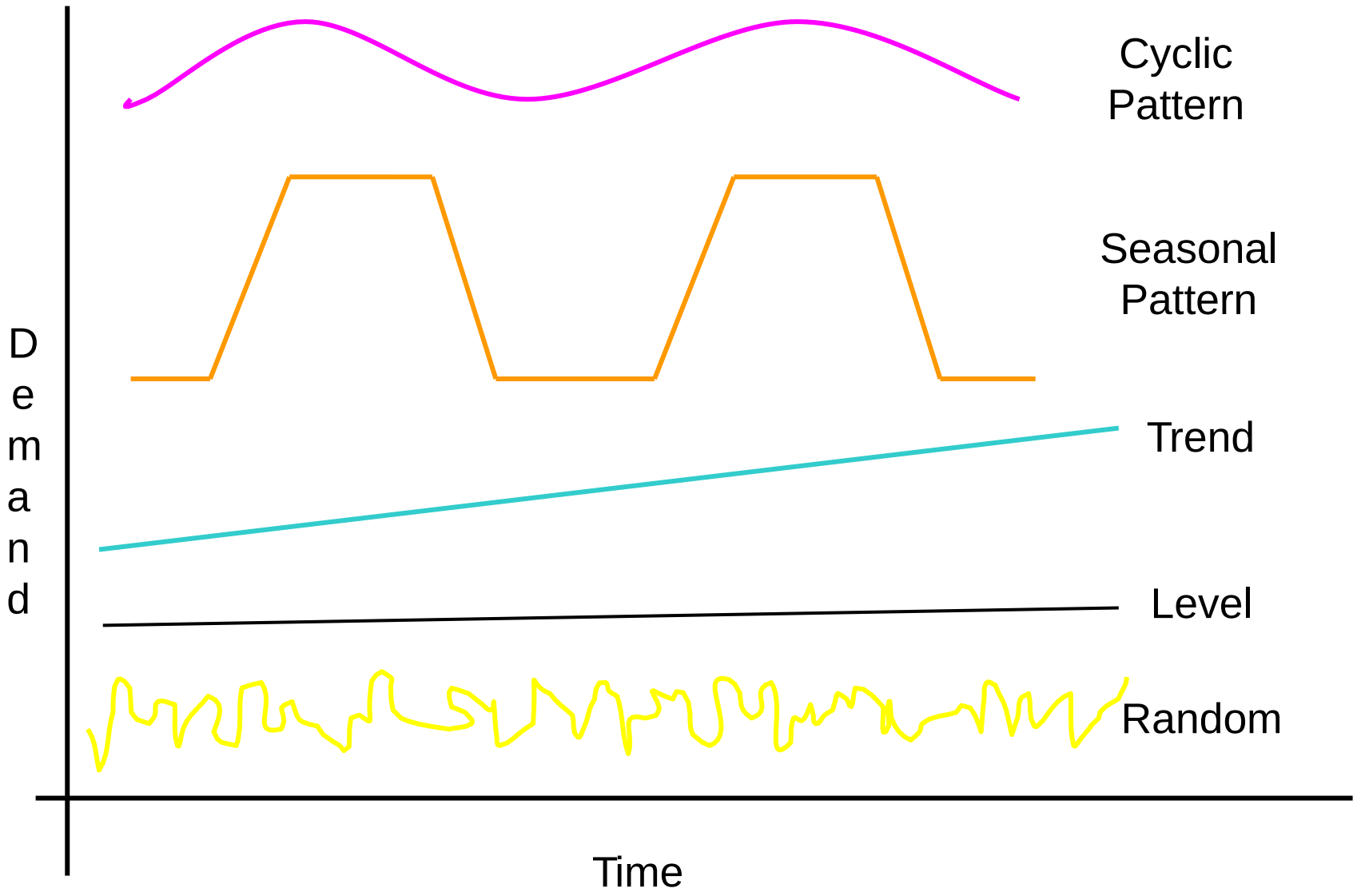
Combination

- Use other information to stratify methods.

e.g.:

  - Impact of advertisement
  - New competition
  - Obsolescence
  - General economy , etc.

Various Components of Forecasting



Seasonal Forecasting

Only Seasonality:

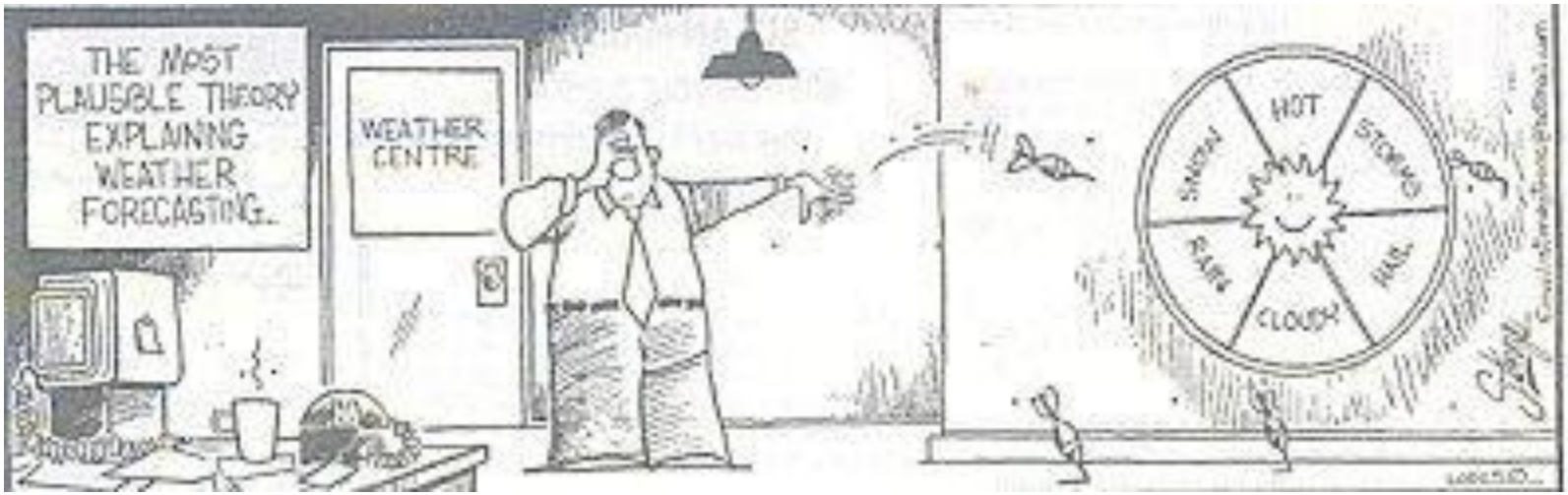
$$\hat{Y}(t) = A + B\sin\left(\frac{2\pi}{p}t\right) + C\cos\left(\frac{2\pi}{p}t\right)$$

Seasonality with Trend:

$$\hat{Y}(t) = A + Bt + C\sin\left(\frac{2\pi}{p}t\right) + D\cos\left(\frac{2\pi}{p}t\right)$$

Here, p is the no. of time periods in a season.

Methods of Forecasting


THE MOST PLAUSIBLE THEORY EXPLAINING WEATHER FORECASTING...
WEATHER CENTRE
HOT
STORMY
SNOW
RAIN
CLOUDY

ECONOMIC FORECAST

Methods of forecasting

- Moving Average
  - Simple Moving Average
  - Weighted Moving Average
- Exponential Smoothing
- time-series - Method of Least Squares

Simple Moving Average

- An averaging period "n" is selected.

- The forecast for the next period ($t + 1$) is the arithmetic average of the most recent actual demands over n. (Here, $t \geq n$).

$$F_{t+1} = \frac{1}{n} \sum_{i=t-n+1}^{t} X_i$$

- It is called a "simple" average because each observation used to compute the average is equally weighted.

Simple Moving Average

- It is called “moving” because, as demand in period $t + 1$ becomes available, the demand in period $t + 1 - n$ is no longer used to make the forecast.

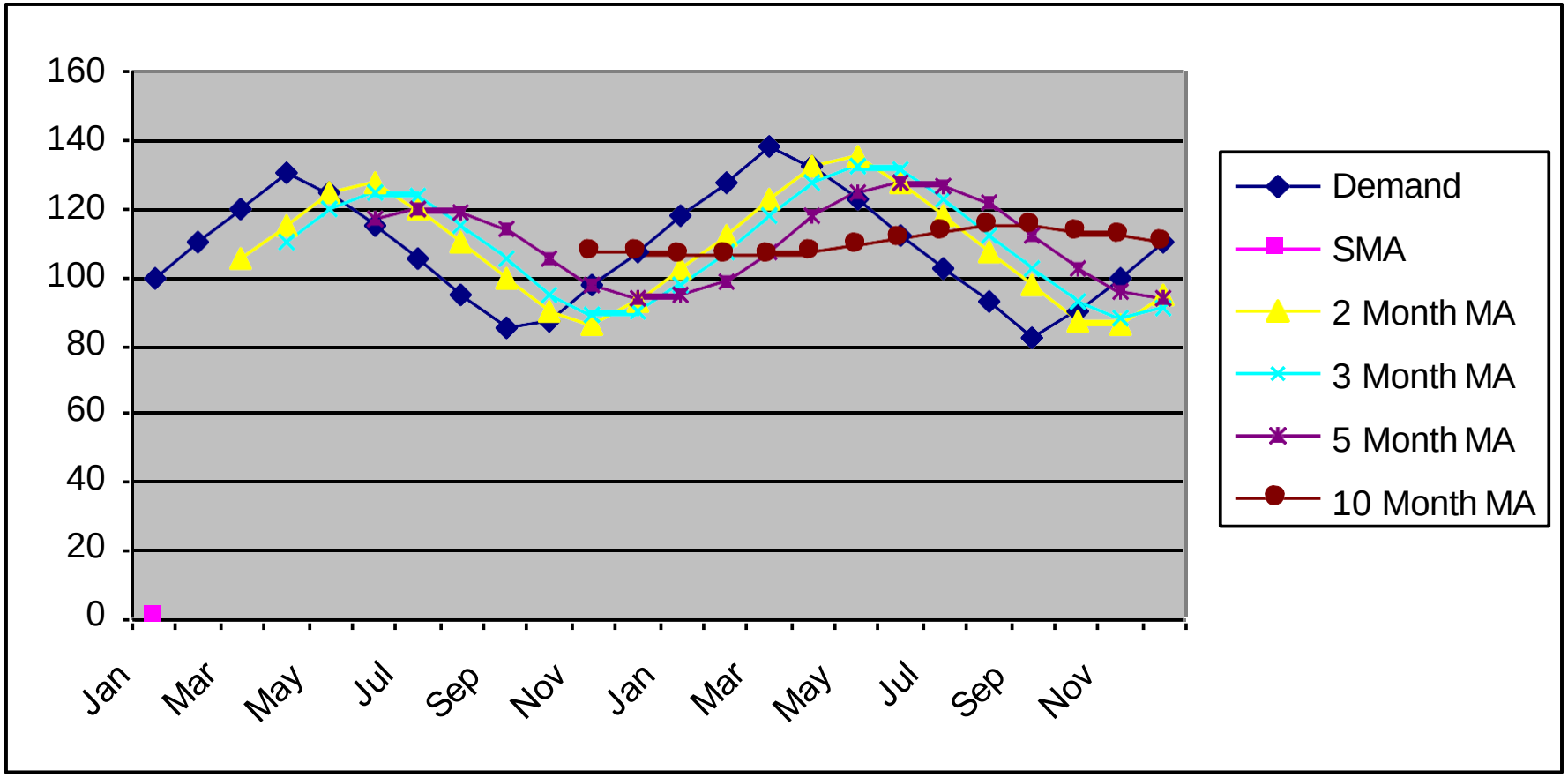
- By increasing n, the forecast becomes less responsive to fluctuations in demand (low impulse response).

- By decreasing n, the forecast becomes more responsive to fluctuations in demand (high impulse response).

Understanding Simple Moving Average

		Simple Moving Average			
Month	Demand	2 Month	3 Month	5 Month	10 Month
Jan	100				
Feb	110				
Mar	120	105			
Apr	130	115	110.00		
May	125	125	120.00		
Jun	115	127.5	125.00	117	
Jul	105	120	123.33	120	
Aug	95	110	115.00	119	
Sep	85	100	105.00	114	
Oct	87.5	90	95.00	105	
Nov	97.5	86.25	89.17	97.5	107.25
Dec	107.5	92.5	90.00	94	107
Jan	117.5	102.5	97.50	94.5	106.75
Feb	127.5	112.5	107.50	99	106.5
Mar	137.5	122.5	117.50	107.5	106.25
Apr	132.5	132.5	127.50	117.5	107.5
May	122.5	135	132.50	124.5	109.25
Jun	112.5	127.5	130.83	127.5	111
Jul	102.5	117.5	122.50	126.5	112.75

Understanding Simple Moving Average



Weighted Moving Average

- This is a variation on the simple moving average.
- Weights used to compute the average are not equal.
- This allows more recent demand data to have a greater effect on the moving average and, therefore, the forecast.

$$F_{t+1} = \frac{\sum_{i=t-n+1}^{t} W_i X_i}{\sum_{i=t-n+1}^{t} W_i}$$

Weighted Moving Average

- The weights can add to 1 and, in general, demands in more recent periods are given higher weightage.

- The distribution of the weights determine impulse response of the forecast.

So, in summary, the Moving Average method has the following properties:

- It smoothens the curve.
- It is independent of relationship between Y and X.
- Only the last n observations are considered to make the forecast for the next period.
- It lags in trend.
- Forecast can be made only for next immediate period.

Exponential Smoothing

- The weights used to compute the forecast are exponential.

- The forecast is the sum of the old forecast and a portion of the forecast error.

Simple Exponential Smoothing Method

- The exponential smoothing equation is

$$F_{t+1} = \alpha y_t + (1 - \alpha) F_t$$

  - $\boldsymbol{F_{t+1}}$ **= forecast for the next period** $(\boldsymbol{t} + \mathbf{1})$
  - $\boldsymbol{\alpha}$ **= smoothing constant**
  - $\boldsymbol{y_t}$ **= actual demand in period** $\boldsymbol{t}$
  - $\boldsymbol{F_t}$ **= forecast made for period** $\boldsymbol{t}$**.**

- The forecast F_{t+1} is made by assigning a weight of α to the most recent observation y_t and a weight of $1 - \alpha$ to the most recent forecast F_t.

$$F_{t+1} = \alpha y_t + (1-\alpha)F_t$$
$$F_{t+1} = \alpha y_t + (1-\alpha)(\alpha y_{t-1} + (1-\alpha)F_{t-1})$$
$$F_{t+1} = \alpha y_t + (1-\alpha)\alpha y_{t-1} + (1-\alpha)^2 F_{t-1}$$

Similarly we keep on replacing forecast terms (F) with demand terms (y),

$$F_{t+1} = \alpha y_t + (1-\alpha)\alpha y_{t-1} + (1-\alpha)^2 \alpha y_{t-2} + \cdots + (1-\alpha)^{t-n}\alpha y_{t-n} + \underline{(1-\alpha)^{t-1}F_2}$$

We can initialize $F_2 = y_1$, i.e., demand in the 1st period is taken as the forecast for the 2nd period.

$$\boxed{F_{t+1} = \alpha y_t + (1-\alpha)\alpha y_{t-1} + (1-\alpha)^2 \alpha y_{t-2} + \cdots + (1-\alpha)^{t-n}\alpha y_{t-n} + (1-\alpha)^{t-1} y_1}$$

n	Sum of Coefficients			
	Alpha			
	0.1	0.2	0.3	0.4
1	0.1000	0.2000	0.3000	0.4000
2	0.1900	0.3600	0.5100	0.6400
3	0.2710	0.4880	0.6570	0.7840
4	0.3439	0.5904	0.7599	0.8704
5	0.4095	0.6723	0.8319	0.9222
6	0.4686	0.7379	0.8824	0.9533
7	0.5217	0.7903	0.9176	0.9720
8	0.5695	0.8322	0.9424	0.9832
9	0.6126	0.8658	0.9596	0.9899
10	0.6513	0.8926	0.9718	0.9940
11	0.6862	0.9141	0.9802	0.9964
12	0.7176	0.9313	0.9862	0.9978
13	0.7458	0.9450	0.9903	0.9987
14	0.7712	0.9560	0.9932	0.9992
15	0.7941	0.9648	0.9953	0.9995
16	0.8147	0.9719	0.9967	0.9997
17	0.8332	0.9775	0.9977	0.9998
18	0.8499	0.9820	0.9984	0.9999
19	0.8649	0.9856	0.9989	0.9999
20	0.8784	0.9885	0.9992	1.0000
21	0.8906	0.9908	0.9994	1.0000
22	0.9015	0.9926	0.9996	1.0000
23	0.9114	0.9941	0.9997	1.0000
24	0.9202	0.9953	0.9998	1.0000
25	0.9282	0.9962	0.9999	1.0000
26	0.9354	0.9970	0.9999	1.0000
27	0.9419	0.9976	0.9999	1.0000
28	0.9477	0.9981	1.0000	1.0000
29	0.9529	0.9985	1.0000	1.0000
30	0.9576	0.9988	1.0000	1.0000

We can notice that, as more and more previous demand points are considered, **the sum of coefficients approaches 1.**

Example - 1

A hospital wishes to predict the monthly requirements for drug and surgical dressing inventory. The actual demand for one such item is shown in the below given table. Develop a forecasting model using exponential smoothing.

Month	24	25	26	27	28	29	30	31	32
Demand (X)	78	65	90	71	80	101	84	60	73

Solution

- To calculate the forecast for period $t + 1$, demand in period t and forecast for period t are needed.

- Demand in the 1st period (month 24) is taken as the forecast for the 2nd period (month 25), i.e., $F_2 = y_1 = 78$.

- Here, the calculations for 4 different values of the smoothing constant α are shown.

Solution

Month	Demand	Forecast			
		Alpha 0.1	Alpha 0.2	Alpha 0.3	Alpha 0.4
24	78				
25	65	78	78	78	78
26	90	77	75	74	73
27	71	78	78	79	80
28	80	77	77	77	76
29	101	77	78	78	78
30	84	79	83	85	87
31	60	80	83	85	86
32	73	78	78	78	76
33		78	77	77	75

Solution

- In practice, the value of α is fixed in such a way so as to minimize the given measure of forecast error.

- Forecast for month 33, i.e., period 10 at $\alpha = 0.2$ is calculated as follows:

$$F_{10} = \alpha y_9 + (1 - \alpha) F_9$$

$$F_{10} = (0.2)(73) + (1 - 0.2)(78)$$

$$\boxed{F_{10} = 77 \text{ units}}$$

Solution

Month	Demand	Forecast			
		Alpha 0.1	Alpha 0.2	Alpha 0.3	Alpha 0.4
24	78				
25	65	78	78	78	78
26	90	77	75	74	73
27	71	78	78	79	80
28	80	77	77	77	76
29	101	77	78	78	78
30	84	79	83	85	87
31	60	80	83	85	86
32	73	78	78	78	76
33		78	77	77	75

For the same exponential smoothing problem, the errors values are calculated as follows:

Forecast Errors Example

Month (t)	Demand	Forecast F_t at $\alpha = 0.1$	Error	Abs. Error	% Error	% Abs Error	MAD_t	TS_t
24	78							
25	65	78	-13	13	-0.20	0.20	13.00	-1.00
26	90	77	13	13	0.15	0.15	13.15	0.02
27	71	78	-7	7	-0.10	0.10	11.11	-0.61
28	80	77	3	3	0.03	0.03	9.00	-0.45
29	101	78	23	23	0.23	0.23	11.88	1.63
30	84	80	4	4	0.05	0.05	10.58	2.21
31	60	80	-20	20	-0.34	0.34	11.97	0.26
32	73	78	-5	5	-0.07	0.07	11.14	-0.20

MAD_t- Mean Absolute Deviation up to period t
TS_t- Tracking Signal up to period t

Forecast Errors Summary

α	0.1	0.2	0.3	0.4
Sum of Errors	-2	-5	-7	-9
MPE	-3.12%	-3.65%	-4.10%	-4.42%
MAD	11.14	11.37	11.60	12.16
MAPE	14.65%	15.05%	15.41%	16.15%
RMSE	13.270	13.855	14.399	14.910
TS	-0.2006	-0.4255	-0.6143	-0.7204

MPE- Mean Percentage Error
MAD- Mean Absolute Deviation
MAPE-Mean Absolute Percentage Error
RMSE- Root Mean Square Error
TS- Tracking Signal

Forecast Errors

Use

- To set safety stocks or safety capacity
  - To ensure a desired level of protection against stock out
- To monitor erratic demand observations or outliers
  - Should be carefully evaluated
  - Perhaps rejected from the data
- To determine when the forecasting method is no longer tracking actual demand and needs to be reset
- Monitor validity of forecasts

Notations

- D_t = Demand during period t
- F_{t+1} = Forecast demand for period $t + 1$
- D_t- F_t = Forecast error (e_t) in period t

Cumulative sum of Forecast Errors (CFE)

$$CFE = \sum_{t=1}^{n} e_t$$

Where, $e_t = D_t - F_t$

Mean Percentage Error (MPE)

$$MPE = \frac{1}{n}\sum_{t=1}^{n}\frac{e_t}{D_t}$$

Mean Absolute Deviation of forecast errors (MAD)

$$MAD = \frac{1}{n}\sum_{t=1}^{n}|e_t|$$

Mean Squared Error (MSE) & Root Mean Squared Error (RMSE)

$$MSE = \frac{1}{n}\sum_{t=1}^{n} e_t^2$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n} e_t^2}$$

Mean Absolute Percentage Error (MAPE)

$$MAPE = \frac{\sum_{t=1}^{n} \left| \frac{e_t}{D_t} \right| 100}{n}$$

(expressed as a percentage)

Tracking Signal (TS)
(most recent)

TS = Cumulative sum of forecast deviation / MAD

$$TS = CFE / MAD$$

[Error in forecast : $e_t = D_t - F_t$]

A tracking signal monitors any forecasts that have been made in comparison with actuals, and warns when there are unexpected **departures of the outcomes from the forecasts.**

A good or a better tracking signal denotes **the one with less cumulative errors**

Holt's Exponential smoothing

- Holt's two parameter exponential smoothing method is an extension of simple exponential smoothing.
- It adds a growth factor (or trend factor) to the smoothing equation as a way of adjusting for the trend.
- Three equations and two smoothing constants are used in the model.
  - The exponentially smoothed series or current level estimate.

$$L_t = \alpha Y_t + (1-\alpha)(L_{t-1} + b_{t-1})$$

  - The trend estimate.

$$b_t = \beta(L_t - L_{t-1}) + (1-\beta)b_{t-1}$$

  - Forecast *m* periods into the future.

$$F_{t+m} = L_t + mb_t$$

  - L_t = Estimate of the level of the series at time t
  - α = smoothing constant for the data.
  - y_t = new observation or actual value of series in period t.
  - β = smoothing constant for trend estimate
  - b_t = estimate of the slope of the series at time t
  - m = periods to be forecast into the future.

Holt's Exponential smoothing

- The weight α and β can be selected subjectively or by minimizing a measure of forecast error.
- Large weights result in more rapid changes in the component.
- Small weights result in less rapid changes.
- The initialization process for Holt's linear exponential smoothing requires two estimates:
  - One to get the first smoothed value for L_1
  - The other to get the trend b_1.
- One alternative is to set $L_1 = y_1$ and

$$b_1 = y_2 - y_1 \qquad or$$

$$b_1 = \frac{y_4 - y_1}{3} \qquad or$$

$$b_1 = 0$$

Example: Quarterly sales of a company

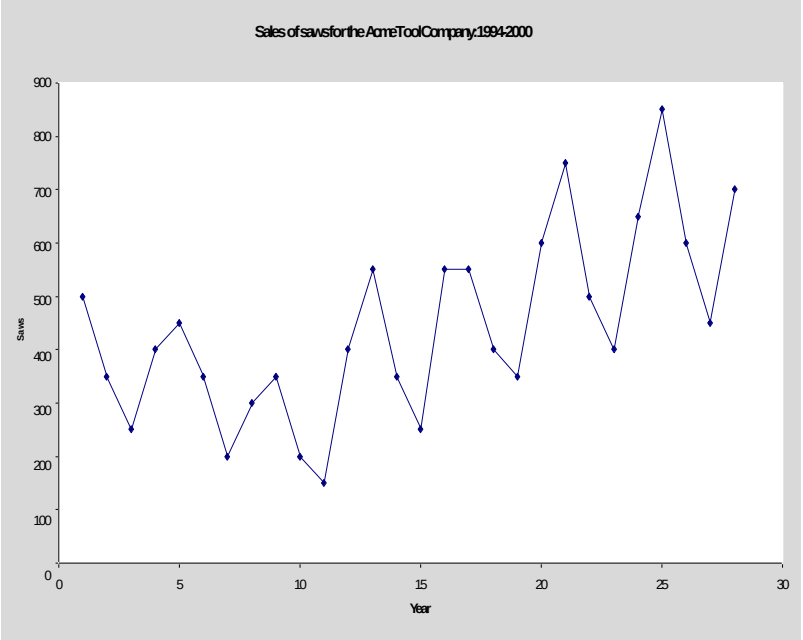
Year	Quarter	sales
2014	1	500
	2	350
	3	250
	4	400
2015	1	450
	2	350
	3	200
	4	300
2016	1	350
	2	200
	3	150
	4	400
2017	1	550
	2	350
	3	250
	4	550
2018	1	550
	2	400
	3	350
	4	600
2019	1	750
	2	500
	3	400
	4	650
2020	1	850
	2	600
	3	450
	4	700

Example: Quarterly sales of a company

- Examination of the plot shows:

  A non-stationary time series data.

  - Seasonal variation seems to exist.
    - **Sales for the first and fourth quarter are larger than other quarters.**



Example: Holt's model to produce forecasts

- We need two initial values
  - The first smoothed value for L_1
  - The initial trend value b_1.
- We will use the first observation for the estimate of the smoothed value L_1, and the initial trend value $b_1 = 0$.
- We will use $\alpha = .3$ and $\beta = .1$.

Year	Quarter	t	sales	L_t	b_t	F_{t+m}
2014	1	1	500	500.00	0.00	500.00
	2	2	350	455.00	-4.50	500.00
	3	3	250	390.35	-10.52	450.50
	4	4	400	385.88	-9.91	379.84
2015	1	5	450	398.18	-7.69	375.97
	2	6	350	378.34	-8.90	390.49
	3	7	200	318.61	-13.99	369.44
	4	8	300	303.23	-14.13	304.62
2016	1	9	350	307.38	-12.30	289.11
	2	10	200	266.55	-15.15	295.08
	3	11	150	220.98	-18.19	251.40
	4	12	400	261.95	-12.28	202.79
2017	1	13	550	339.77	-3.27	249.67
	2	14	350	340.55	-2.86	336.50
	3	15	250	311.38	-5.49	337.69
	4	16	550	379.12	1.83	305.89
2018	1	17	550	431.67	6.90	380.95
	2	18	400	427.00	5.74	438.57
	3	19	350	407.92	3.26	432.74
	4	20	600	467.83	8.93	411.18
2019	1	21	750	558.73	17.12	476.75
	2	22	500	553.10	14.85	575.85
	3	23	400	517.56	9.81	567.94
	4	24	650	564.16	13.49	527.37
2020	1	25	850	659.35	21.66	577.65
	2	26	600	656.71	19.23	681.01
	3	27	450	608.16	12.45	675.94
	4	28	700	644.43	14.83	620.61

- Level Estimate, $L_t = \alpha Y_t + (1-\alpha)(L_{t-1} + b_{t-1})$

$$L_2 = 0.3 * 350 + (1-0.3)(500-0) = \boxed{\mathbf{455}}$$

- Trend Estimate, $b_t = \beta(L_t - L_{t-1}) + (1-\beta)b_{t-1}$

$$b_2 = 0.1 * (455 - 500) + (1-0.1) * 0 = \boxed{\mathbf{-4.5}}$$

- Forecast made at the end of period t for period $t+m$,

$$F_{t+m} = L_t + mb_t$$

- Forecast made at the end of period 2 for period 3 (At $t = 2$, $m = 1$)

$$F_3 = F_{2+1} = L_2 + 1 * b_2 = 455 + 1 * (-4.5) = \boxed{\mathbf{450.5}}$$

Seasonal Forecasting : Example

Seasonal data with 5 seasonals(Days) across the period (Week)

1	8	4	2	6	15	35	7
2	9	5	4	7	15	40	8
3	11	5	4	8	17	45	9
4	13	6	5	10	16	50	10
Weekly Total	41	20	15	31	63	-	-

Seasonal Index Computation based on based on 'Daily Average'

Week	M	T	W	TH	F	Daily Total
1	1.14	0.57	0.29	0.86	2.14	5
2	1.13	0.63	0.50	0.88	1.88	5.02
3	1.22	0.56	0.44	0.89	1.89	5
4	1.30	0.60	0.50	1.00	1.60	5
Total	4.79	2.36	1.73	3.63	7.51	
Average	1.20	0.59	0.43	0.91	1.88	5.01

- **Data / Daily Average**
- **For e.g., Monday Week 1 = 8/7 = 1.14**
  Monday Week 2 = 9/8 = 1.13

Deseasonalized data (unadjusted)

Week	M	T	W	TH	F	Daily Total	Daily Average
1	6.67	6.78	4.65	6.59	7.98	32.67	6.52
2	7.5	8.47	9.3	7.69	7.98	40.94	8.19
3	9.17	8.47	9.3	8.79	9.04	44.77	8.95
4	10.83	10.17	11.63	10.99	8.51	52.13	10.43

- **Data / Average on each day**
- **For e.g. Monday Week 1 = 8 / 1.2 = 6.67**
  Monday Week 2 = 9 / 1.2 = 7.5

Forecast using regression analysis on Deseasonalized data

$$y(t) = f_d(5.4 + 1.25t)$$

$Where\ f_d$ - **Daily seasonal Index**

Week *t*	M	T	W	TH	F	Daily Average
5	13.98	6.87	5.01	10.60	21.90	11.65
6	15.48	7.61	5.55	11.74	24.25	12.90

- **For e.g. Monday Week 5 = 1.2 (5.4+1.25 ×5)**

Winter's Exponential Smoothing

- Winter's exponential smoothing model is the second extension of the basic Exponential smoothing model.
- It is used for data that exhibit both trend and seasonality.
- It is a three parameter model that is an extension of Holt's method.
- An additional equation adjusts the model for the seasonal component.

Multiplicative Seasonality

- The four equations necessary for Winter's multiplicative method are:
  - The exponentially smoothed series:

$$L_t = \alpha \frac{y_t}{S_{t-s}} + (1-\alpha)(L_{t-1} + b_{t-1})$$

  - The trend estimate:

$$b_t = \beta(L_t - L_{t-1}) + (1-\beta)b_{t-1}$$

  - The seasonality estimate:

$$S_t = \gamma \frac{y_t}{L_t} + (1-\gamma)S_{t-s}$$

Winter's Exponential Smoothing

- Forecast *m* period into the future:

$$F_{t+m} = (L_t + mb_t)\, S_{t+m-s}$$

- **L_t = level of series.**
- **α = smoothing constant for the data.**
- **y_t = new observation or actual value in period t.**
- **β = smoothing constant for trend estimate.**
- **b_t = trend estimate.**
- **γ = smoothing constant for seasonality estimate.**
- **S_t =seasonal component estimate.**
- **m = Number of periods in the forecast lead period.**
- **s = length of seasonality (number of periods in the season)**

- F_{t+m} **= forecast for m periods into the future.**

Winter's Exponential Smoothing

- As with Holt's linear exponential smoothing, the weights α, β, and γ can be selected subjectively or by minimizing a measure of forecast error.
- As with all exponential smoothing methods, we need initial values for the components to start the algorithm.
- To start the algorithm, the initial values for L_t, the trend b_t, and the indices S_t must be set.

Winter's Exponential Smoothing

- To determine initial estimates of the seasonal indices we need to use at least one complete season's data (i.e. s periods).Therefore,we initialize trend and level at period s.
- Initialize level as:

$$L_s = \frac{1}{s}(y_1 + y_2 + \ldots + y_s)$$

- Initialize trend as

$$b_s = \frac{1}{s}\left(\frac{y_{s+1} - y_1}{s} + \frac{y_{s+2} - y_2}{s} + \ldots \frac{y_{s+s} - y_s}{s}\right)$$

- Initialize seasonal indices as:

$$S_1 = \frac{y_1}{L_s}, S_2 = \frac{y_2}{L_s}, \quad S_s = \frac{y_s}{L_s}$$

Example

(α = 0.4, β= 0.1, γ = 0.3 ,s =4, m=1, unless otherwise stated)

Year	Quarter	t	Sales	L_t	b_t	S_t	F_{t+m}
2014	1	1	500			1.333	
	2	2	350			0.933	
	3	3	250			0.667	
	4	4	400	375	-12.50	1.067	
2015	1	5	450	352.50	-13.50	1.316	483.33
	2	6	350	353.40	-12.06	0.950	316.40
	3	7	200	324.80	-13.71	0.651	227.56
	4	8	300	299.15	-14.91	1.048	331.83
2016	1	9	350	276.91	-15.64	1.301	374.16
	2	10	200	240.93	-17.67	0.914	248.32
	3	11	150	226.06	-17.39	0.655	145.43
	4	12	400	277.94	-10.47	1.165	218.58
2017	1	13	550	329.64	-4.25	1.411	347.88
	2	14	350	348.35	-1.95	0.941	297.52
	3	15	250	360.50	-0.54	0.667	226.90
	4	16	550	404.81	3.94	1.223	419.35

Year	Quarter	t	Sales	L_t	b_t	S_t	F_{t+m}	
2018	1	17	550	401.17	3.18	1.399	576.74	
	2	18	400	412.56	4.00	0.950	380.69	
	3	19	350	459.97	8.34	0.695	277.67	
	4	20	600	477.21	9.23	1.233	572.79	
2019	1	21	750	506.31	11.22	1.424	680.53	
	2	22	500	521.07	11.57	0.953	491.60	
	3	23	400	549.84	13.29	0.705	370.12	
	4	24	650	548.69	11.85	1.219	694.55	
2020	1	25	850	575.14	13.31	1.440	798.03	
	2	26	600	604.96	14.96	0.964	560.67	
	3	27	450	627.40	15.71	0.708	436.83	
	4	28	700	615.61	12.96	1.194	783.78	m
2021	1	29					905.10	1
	2	30					612.34	2
	3	31					460.37	3
	4	32					793.94	4

Calculations

- Initialization of Level (L_s) and Trend (b_s) (Here, $s = 4$):

$$L_4 = \frac{y_1 + y_2 + y_3 + y_4}{4} = \frac{500 + 350 + 250 + 400}{4}$$

$$\boxed{\boldsymbol{L_4 = 375}}$$

$$b_4 = \frac{1}{4}\left(\frac{(y_5 - y_1)}{4} + \frac{(y_6 - y_2)}{4} + \frac{(y_7 - y_3)}{4} + \frac{(y_8 - y_4)}{4}\right)$$

$$b_4 = \frac{1}{4}\left(\frac{(450 - 500)}{4} + \frac{(350 - 350)}{4} + \frac{(200 - 250)}{4} + \frac{(300 - 400)}{4}\right)$$

$$\boxed{\boldsymbol{b_4 = -12.5}}$$

Calculations

- Initialization of Seasonality estimates:

$$S_1 = \frac{y_1}{L_4} = \frac{500}{375} = \boxed{\mathbf{1.333}}$$

$$S_2 = \frac{y_2}{L_4} = \frac{350}{375} = \boxed{\mathbf{0.933}}$$

$$S_3 = \frac{y_3}{L_4} = \frac{250}{375} = \boxed{\mathbf{0.667}}$$

$$S_4 = \frac{y_4}{L_4} = \frac{400}{375} = \boxed{\mathbf{1.067}}$$

Calculations

- Forecast made at the end of period 4 for period 5 (F_{t+m}): (Here, $t = 4, m = 1$)

$$F_5 = F_{4+1} = (L_4 + 1 * b_4)S_1$$

$$F_5 = (375 + 1 * (-12.5))1.333 = \boxed{\mathbf{483.33}}$$

- Level estimate for period 5 (L_5):

$$L_5 = \alpha \frac{y_5}{S_1} + (1 - \alpha)(L_4 + b_4)$$

$$L_5 = 0.4 \frac{450}{1.333} + (1 - 0.4)\big(375 + (-12.5)\big) = \boxed{\mathbf{352.5}}$$

Calculations

- Seasonality estimate for period 5 (S_5):

$$S_5 = \gamma \frac{y_5}{L_5} + (1-\gamma)S_1$$

$$S_5 = 0.3\frac{450}{352.5} + (1-0.3)1.333 = \boxed{\mathbf{1.316}}$$

- Trend estimate for period 5 (b_5):

$$b_5 = \beta(L_5 - L_4) + (1-\beta)b_4$$

$$b_5 = 0.1(352.5 - 375) + (1-0.1)(-12.5) = \boxed{\mathbf{-13.5}}$$

Calculations

- Forecast made at the end of period 5 for period 6 (F_{t+m}): (Here, $t = 5, m = 1$)

$$F_6 = F_{5+1} = (L_5 + 1 * b_5)S_2$$

$$F_6 = (352.5 + 1 * (-13.5))0.933$$

$$\boxed{F_6 = \mathbf{316.4}}$$

Additive Seasonality

- The seasonal component in Holt-Winters' method.
- The basic equations for Holt's Winters' additive method are:

$$L_t = \alpha(y_t - S_{t-s}) + (1-\alpha)(L_{t-1} + b_{t-1})$$

$$b_t = \beta(L_t - L_{t-1}) + (1-\beta)b_{t-1}$$

$$S_t = \gamma(y_t - L_t) + (1-\gamma)S_{t-s}$$

$$F_{t+m} = L_t + b_t m + S_{t+m-s}$$

Additive Seasonality

- The initial values for L_s and b_s are identical to those for the multiplicative method.
- To initialize the seasonal indices we use

$$S_1 = y_1 - L_s, \quad S_2 = y_2 - L_s, \quad S_s = y_s - L_s$$

Example

(α = 0.4, β= 0.1, γ = 0.3 ,s =4,
m = 1, unless otherwise stated)

Year	Quarter	t	Sales	L_t	b_t	S_t	F_{t+m}
2014	1	1	500			125	
	2	2	350			-25	
	3	3	250			-125	
	4	4	400	375	-12.50	25	
2015	1	5	450	347.50	-14.00	118.25	487.50
	2	6	350	350.10	-12.34	-17.53	308.50
	3	7	200	332.66	-12.85	-127.30	212.76
	4	8	300	301.88	-14.64	16.93	344.81
2016	1	9	350	265.04	-16.86	108.26	405.49
	2	10	200	235.92	-18.09	-23.05	230.65
	3	11	150	241.62	-15.71	-116.59	90.54
	4	12	400	288.77	-9.42	45.22	242.84
2017	1	13	550	344.30	-2.93	137.49	387.61
	2	14	350	354.04	-1.66	-17.35	318.33
	3	15	250	358.07	-1.09	-114.04	235.79
	4	16	550	416.10	4.82	71.83	402.20

Year	Quarter	t	Sales	L_t	b_t	S_t	F_{t+m}	
2018	1	17	550	417.55	4.48	135.98	558.41	
	2	18	400	420.16	4.30	-18.19	404.69	
	3	19	350	440.29	5.88	-106.91	310.42	
	4	20	600	478.97	9.16	86.59	517.99	
2019	1	21	750	538.49	14.19	158.64	624.11	
	2	22	500	538.88	12.82	-24.40	534.49	
	3	23	400	533.78	11.02	-114.97	444.79	
	4	24	650	552.25	11.77	89.94	631.40	
2020	1	25	850	614.95	16.86	181.56	722.66	
	2	26	600	628.85	16.56	-25.73	607.42	
	3	27	450	613.24	13.35	-129.45	530.44	
	4	28	700	619.98	12.69	86.96	716.52	m
2021	1	29					814.22	1
	2	30					636.24	2
	3	31					549.09	3
	4	32					782.07	4

Calculations

- Initialization of Level (L_s) and Trend (b_s) (Here, $s = 4$):

$$L_4 = \frac{y_1 + y_2 + y_3 + y_4}{4} = \frac{500 + 350 + 250 + 400}{4}$$

$$\boxed{\boldsymbol{L_4 = 375}}$$

$$b_4 = \frac{1}{4}\left(\frac{(y_5 - y_1)}{4} + \frac{(y_6 - y_2)}{4} + \frac{(y_7 - y_3)}{4} + \frac{(y_8 - y_4)}{4}\right)$$

$$b_4 = \frac{1}{4}\left(\frac{(450 - 500)}{4} + \frac{(350 - 350)}{4} + \frac{(200 - 250)}{4} + \frac{(300 - 400)}{4}\right)$$

$$\boxed{\boldsymbol{b_4 = -12.5}}$$

Calculations

- Initialization of Seasonality estimates:

$$S_1 = y_1 - L_4 = 500 - 375 = \boxed{\mathbf{125}}$$

$$S_2 = y_2 - L_4 = 350 - 375 = \boxed{\mathbf{-25}}$$

$$S_3 = y_3 - L_4 = 250 - 375 = \boxed{\mathbf{-125}}$$

$$S_4 = y_4 - L_4 = 400 - 375 = \boxed{\mathbf{25}}$$

Calculations

- Forecast made at the end of period 4 for period 5 (F_{t+m}): (Here, $t = 4, m = 1$)

$$F_5 = F_{4+1} = L_4 + 1 * b_4 + S_1$$

$$F_5 = 375 + 1 * (-12.5) + 125 = \boxed{\mathbf{487.50}}$$

- Level estimate for period 5 (L_5):

$$L_5 = \alpha(y_5 - S_1) + (1 - \alpha)(L_4 + b_4)$$

$$L_5 = 0.4(450 - 125) + (1 - 0.4)\big(375 + (-12.5)\big) = \boxed{\mathbf{347.5}}$$

Calculations

- Seasonality estimate for period 5 (S_5):

$$S_5 = \gamma(y_5 - L_5) + (1 - \gamma)S_1$$

$$S_5 = 0.3(450 - 347.5) + (1 - 0.3)125 = \boxed{\mathbf{118.25}}$$

- Trend estimate for period 5 (b_5):

$$b_5 = \beta(L_5 - L_4) + (1 - \beta)b_4$$

$$b_5 = 0.1(347.5 - 375) + (1 - 0.1)(-12.5) = \boxed{\mathbf{-14}}$$

Calculations

- Forecast made at the end of period 5 for period 6 (F_{t+m}): (Here, $t = 5, m = 1$)

$$F_6 = F_{5+1} = L_5 + 1 * b_5 + S_2$$

$$F_6 = 347.5 + 1 * (-14) + (-25)$$

$$\boxed{F_6 = \mathbf{308.5}}$$

Selection of Forecasting Method

1. User and system sophistication
2. Time and Resources available
3. Use or decision characteristics
4. Data availability (and accuracy)
5. Data pattern (plot on graph and check – I Step)

Selection of Forecasting Method

- Best fit is not necessarily best predictor.
- Do in 2 steps
  - First fit various models,
  - Then choose one with least error in the training set and predict for the test set.

Selection of Forecasting Method

Sometimes, human beings do better than quantitative forecasts !!!

Using Forecast in an Organization

- Bias
  - Due to Sales Boosting
  - Due to production reducing
- Use of probability in forecasting?
- Use of shared forecast?
  - Collective inputs

Regression based Models

- The most general model used in forecasting is the polynomial of the basic form

$$\hat{Y}(t) = a_0 + a_1 t + a_2 t^2 + .. + a_{n-1} t^{n-1} + a_n t^n$$

- **t is called predictor variable and $\hat{Y}$ is called the forecast (response) variable.**

- Truncating later terms, we get

  $\hat{Y}(t) = a_0$: Constant Model

  $\boldsymbol{\hat{Y}(t) = a_0 + a_1 t}$ **: Simple Linear Model with Slope $\boldsymbol{a_1}$**

  $\hat{Y}(t) = a_0 + a_1 t + a_2 t^2$: Quadratic model

- Exponential Model
  - $\hat{Y}(t) = ae^{bt}$

  A logarithmic transformation gives:
  - $\ln\left[\hat{Y}(t)\right] = \ln(a) + \ln(e^{bt})$
  - $\ln[\hat{Y}(t)] = \ln(a) + bt$

Method of least squares

Consider a simple linear regression model.

Let Y_i be the actual sales and $\hat{Y}_i$ be the forecast sales.

Let e_i be the error for the ith period.

$$e_i = Y_i - \hat{Y}_i$$

But, $\hat{Y}_i = a + bX_i$

Therefore, $$e_i = Y_i - a - bX_i$$

Let $E(a,b)$ be the sum of the squared errors.

$$E(a,b) = \sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right)^2 = \sum_{i=1}^{n} (Y_i - a - bX_i)^2$$

Find a, b such that E is minimum.

$$\frac{\partial E}{\partial a} = 0$$

$$\frac{\partial E}{\partial a} = -2 \sum (Y_i - a - b\,X_i) = 0$$

$$\Rightarrow \sum Y_i - na - b\sum X_i = 0$$

$$\Rightarrow \sum Y_i = na + b\sum X_i \text{.....(1)}$$

$$\frac{\partial E}{\partial b} = 0$$

$$\frac{\partial E}{\partial b} = -2\sum_{i=1}^{n}(Y_i - a - bX_i)X_i = 0$$

$$\sum X_i Y_i = a\sum X_i + b\sum X_i^2 \text{.....(2)}$$

$$\sum Y_i = na + b\sum X_i$$

$$\sum X_i Y_i = a\sum X_i + b\sum X_i^2$$

$$= a\sum X_i + \sum X_i\left(\sum Y_i - na\right)$$

$$= a\sum X_i + \sum X_i \sum Y_i - na\sum X_i$$

$$= \sum X_i \sum Y_i + a\left(\sum X_i - n\sum X_i\right)$$

It can be shown that the resulting **least squares estimators b and a** are

$$b = \frac{n\sum_{i=1}^{n} X_i Y_i - \left(\sum_{i=1}^{n} X_i\right)\left(\sum_{i=1}^{n} Y_i\right)}{n\sum_{i=1}^{n} X_i^2 - \left(\sum_{i=1}^{n} X_i\right)^2} = \frac{\sum_{i=1}^{n} X_i Y_i - n\bar{X}\bar{Y}}{\sum_{i=1}^{n} X_i^2 - n\bar{X}^2}$$

$$a = \frac{\sum_{i=1}^{n} Y_i - b\sum_{i=1}^{n} X_i}{n} = \bar{Y} - b\bar{X}$$

The formula for b can also be represented in alternative ways, such as

$$b = \frac{E((X - E(X))(Y - E(Y)))}{E(X - [E(X)])^2} = \frac{E(XY) - E(X)E(Y)}{E(X^2) - [E(X)]^2}$$

$$b = \frac{Cov(X, Y)}{Var(X)}$$

where, $Cov(X, Y)$ is the co-variance of X and Y

If we normalize the scale such that

$$\sum_{i=1}^{n} X_i = 0$$

then

$$b = \frac{\sum_{i=1}^{n} X_i Y_i}{\sum_{i=1}^{n} X_i^2}$$

and

$$a = \frac{\sum_{i=1}^{n} Y_i}{n} = \bar{Y}$$

Correlation Coefficient

- **Correlation (Dependence)**: It is the existence of a statistical (not necessarily causal) relationship between a pair of variables.

- **Pearson Correlation Coefficient ($\boldsymbol{r_{XY}}$)**: It is a relative measure of the degree of **linear** relationship between two variables (X and Y).

$$r_{XY} = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$$

- Where $Cov(X,Y)$ is the co-variance of X and Y, and σ_X and σ_Y are the standard deviations of X and Y respectively.
- r_{XY} always lies in the range $[-1,1]$, with ± 1 indicating perfect correlation and 0 indicating no correlation.

Co-efficient of Determination

- **Coefficient of Determination ($\boldsymbol{R^2}$)**: It is the proportion of total variation in the response that is explained by the regression line.

$$R^2 = \frac{\sum_{i=1}^{n}\left(\hat{Y}_i - \bar{Y}\right)^2}{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}$$

- It lies in the range $[0,1]$, with a value of 0 indicating that the regression line does not explain any variation in the response, and a value of 1 indicating that the regression line fully explains the variations in the response.

Example 2

Sales data for a certain item is given for last 11 years.

Fit regression line and make a forecast for the year 2020.

Taking year 2008 as year zero, the table is constructed as follows:

Year	Demand
2009	2
2010	3
2011	6
2012	10
2013	8
2014	7
2015	12
2016	14
2017	14
2018	18
2019	19

Determination of parameters

Least Squares Fit

Year	X	Y	XY	X^2
2009	1	2	2	1
2010	2	3	6	4
2011	3	6	18	9
2012	4	10	40	16
2013	5	8	40	25
2014	6	7	42	36
2015	7	12	84	49
2016	8	14	112	64
2017	9	14	126	81
2018	10	18	180	100
2019	11	19	209	121
	$\sum_{i=1}^{11} X_i = 66$	$\sum_{i=1}^{11} Y_i = 113$	$\sum_{i=1}^{11} X_i Y_i = 859$	$\sum_{i=1}^{11} X_i^2 = 506$

$$\bar{X} = E(X) = \frac{\sum_{i=1}^{n} X_i}{n} = \frac{66}{11} = \mathbf{6}; \; \bar{Y} = E(Y) = \frac{\sum_{i=1}^{n} Y_i}{n} = \frac{113}{11} = \mathbf{10.273}$$

$$b = \frac{\sum_{i=1}^{n} X_i Y_i - n\bar{X}\bar{Y}}{\sum_{i=1}^{n} X_i^2 - n\bar{X}^2} = \frac{859 - 11 * 6 * 10.273}{506 - 11 * 6^2} = \mathbf{1.645}$$

$$a = \bar{Y} - b\bar{X} = 10.273 - 1.645 * 6 = \mathbf{0.4}$$

Forecasting equation is:

$$\hat{Y} = a + b\,X = \mathbf{0.4} + \mathbf{1.645} * \boldsymbol{X}$$

For year 2020, ($X = 2020 - 2008 = 12$), the forecast is:

$$\hat{Y} = 0.4 + 1.645 * 12 = \mathbf{20.15}\; units$$

The correlation coefficient is given by:

$$r_{XY} = \frac{Cov(X,Y)}{\sigma_X \sigma_Y} = \frac{E(XY) - E(X)E(Y)}{\sqrt{E(X^2) - [E(X)]^2}\sqrt{E(Y^2) - [E(Y)]^2}}$$

$$r_{XY} = \frac{78.091 - 6 * 10.273}{\sqrt{46 - [6]^2}\sqrt{134.818 - [10.273]^2}} = \mathbf{0.9615}$$

The coefficient of determination is given by:

$$R^2 = \frac{\sum_{i=1}^{n}\left(\hat{Y}_i - \bar{Y}\right)^2}{\sum_{i=1}^{n}(Y_i - \bar{Y})^2} = \frac{\sum_{i=1}^{n}(a + bX_i - \bar{Y})^2}{\sum_{i=1}^{n}(Y_i - \bar{Y})^2} = \frac{27.075}{29.289}$$

$$\boldsymbol{R^2 = 0.9244}$$

Regression Analysis

- A simple regression model ($\hat{Y} = a + bX$) developed for making forecasts must suitably answer the following questions:

  - Is the coefficient (b) of the predictor variable X significantly different from 0?

  - How confident can we be of the estimates of the regression coefficient b?

  - How confident can we be that the actual value (Y) of the response variable will lie within a small range of the forecasted value ($\hat{Y}$)?

Regression Analysis

- The first question addresses the significance of the regression coefficient b, specifically whether the true value of b is really different from 0.

- The true value of b is unknown and we use the available sample data to estimate the value of b ($\hat{b}$).

- However, since $\hat{b}$ has been estimated based on a limited number of observations, it is possible that $\hat{b}$ might be non-zero purely by chance.

Regression Analysis

- To test the significance of b, we define the following null hypothesis (H_0) and alternative hypothesis (H_1) as follows:

- $H_0: b = 0; \;\; H_1: b \neq 0$

- Then the standard error of b is computed.

$$SE_b = \frac{\sqrt{\frac{\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2}{n-k-1}}}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2}}$$

- Here, n is the no. of observations, k is the no. of predictors (k=1 for simple regression)

Regression Analysis

- The estimate of b ($\hat{b}$) is divided by this standard error to obtain the test statistic (t).

- This test statistic is then compared with the critical value ($t_{\frac{\alpha}{2},n-2}$) of the Student's t-distribution at a given level of significance (α).

- If t is greater than $t_{\frac{\alpha}{2},n-2}$, the null hypothesis can be rejected and it can be concluded that b is significantly different from zero.

Example

- Considering the regression model for the previous sales example: $\hat{Y} = 0.4 + 1.645X$

$$SE_b = \frac{\sqrt{\frac{24.355}{11-2}}}{\sqrt{110}} = 0.1568$$

$$t = \frac{\hat{b}}{SE_b} = \frac{1.645}{0.1568} = 10.491$$

- At $\alpha = 0.05$,

$$t_c = t_{0.025,9} = 2.262$$

- Since $t > t_c$, we can conclude that b is significantly different from 0.

Regression Analysis

- The second question is concerned with the precision with which the regression coefficient b can be estimated.

- Again, since the true value of b is unknown, we use an estimate of b ($\hat{b}$) calculated based on sample observations to fit the regression equation.

- The probability that the true value of b will be exactly equal to $\hat{b}$ is zero.

Regression Analysis

- However, it is statistically possible to predict (with a degree of confidence) that the true value of b will lie in a small interval around the estimated value $\hat{b}$.

- This interval around the estimated value at a given confidence level C ($C = 1 - \alpha$) is called the confidence interval (CI).

- Lower limit of CI: $LL_b = \hat{b} - t_{\frac{\alpha}{2},n-2} SE_b$
- Upper limit of CI: $UL_b = \hat{b} + t_{\frac{\alpha}{2},n-2} SE_b$

Regression Analysis

- Considering the same example, at $C = 0.95$ ($\alpha = 0.05$)

- Lower limit of CI: $LL_b = 1.645 - 2.262 * 0.1568$
  $LL_b = 1.291$

- Upper limit of CI: $UL_b = 1.645 + 2.262 * 0.1568$
  $UL_b = 2.000$

- Thus we can be 95% certain that the true value of b will lie between 1.291 and 2.000.

Regression Analysis

- The third question is concerned with the precision of the forecasts made using the regression model.

- In other words, we want to know how confident we can be that the true value of the response variable (Y_f) will lie within a small interval around the forecasted value ($\widehat{Y_f} = a + bX_f$) for a given period (f).

- That is, we would like to have a confidence interval around this forecasted value ($\widehat{Y_f}$).

Regression Analysis

- To calculate the confidence interval for $\widehat{Y_f}$, it is first necessary to determine the standard error of forecast (SE_f).

- $SE_f = \left(\sqrt{\frac{\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2}{n-k-1}}\right)\left(\sqrt{1 + \frac{1}{n} + \frac{\left(X_f - \bar{X}\right)^2}{\sum_{i=1}^{n}(X_i - \bar{X})^2}}\right)$

- Where, X_f is the value of the predictor variable at which the forecast is to be made,
- k is the no. of predictor variables ($k = 1$ for simple regression)

Regression Analysis

- If n is large and the values of X_i are not close to the mean $\bar{X}$, the second term in SE_f becomes approximately equal to 1 and can be neglected.

- The confidence interval limits are then calculated as:

- Lower limit of CI: $LL_{Y_f} = \hat{Y}_f - t_{\frac{\alpha}{2},n-2} SE_f$
- Upper limit of CI: $UL_{Y_f} = \hat{Y}_f + t_{\frac{\alpha}{2},n-2} SE_f$

Regression Analysis

- Considering the previous example, in year 2020 $(X_{2020} = 2020 - 2008 = 12)$, SE_{2020} is calculated as:

$$SE_{2020} = \left(\sqrt{\frac{24.355}{11-2}}\right)\left(\sqrt{1 + \frac{1}{11} + \frac{(12-6)^2}{110}}\right) = 1.959$$

- The forecasted value for year 2020 is given by:

$$\hat{Y}_{2020} = \hat{a} + \hat{b}X_{2020} = 0.4 + 1.645 * 12 = 20.15$$

Regression Analysis

- At $C = 0.95$ ($\alpha = 0.05$),
- Lower limit of CI: $LL_{Y_f} = 20.15 - 2.262 * 1.959$

  $LL_{Y_f} = 15.71$

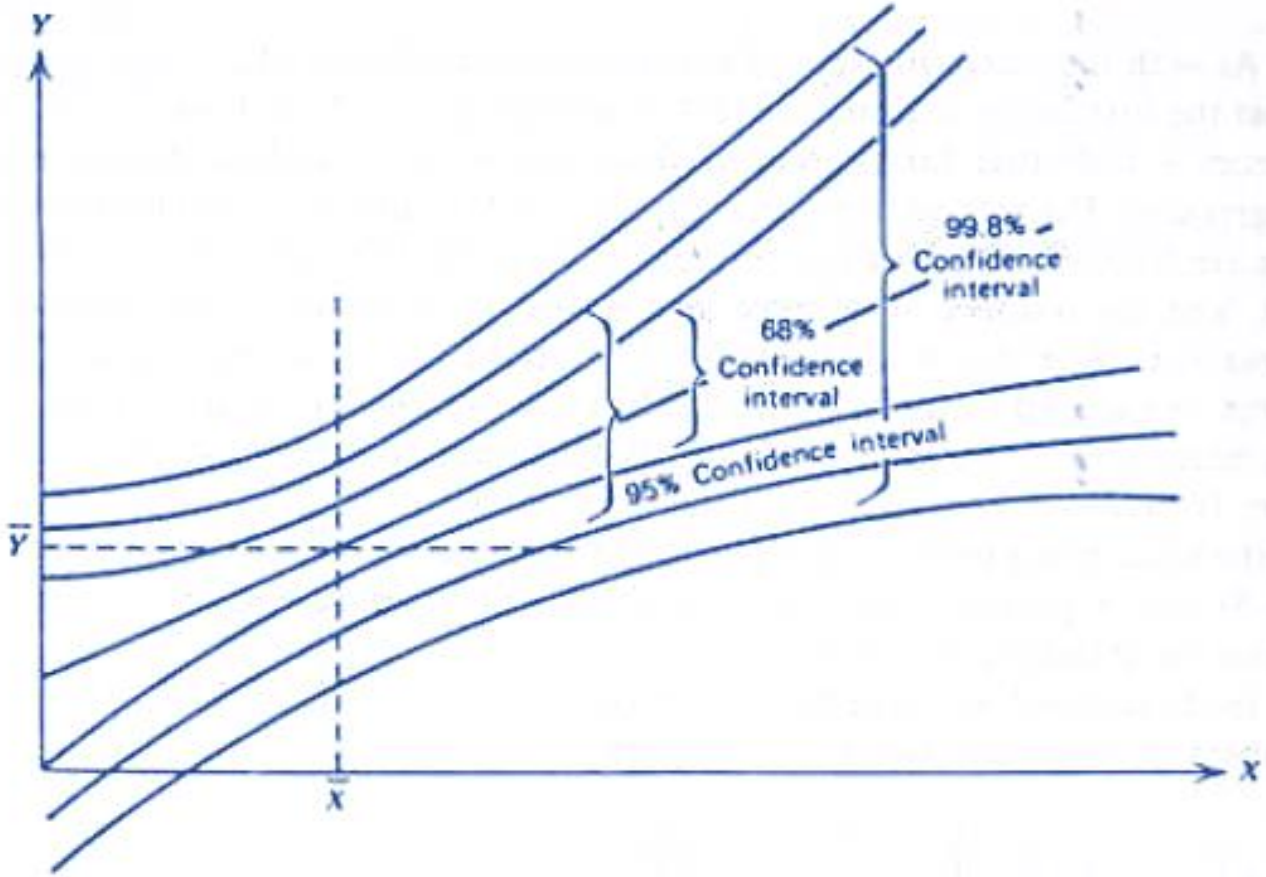
- Upper limit of CI: $UL_{Y_f} = 20.15 + 2.262 * 1.959$

  $UL_{Y_f} = 24.58$

- Thus we can be 95% certain that the actual sales for the year 2020 will lie between 15.71 and 24.58.

Regression Analysis

- The variation in the ranges of confidence of individual forecasts



Variable Transformation in Regression

- In some cases, the response variable may be non-linear in terms of the regression parameters. (e.g. $\hat{Y} = ae^{bt}$)

- The OLS method will not be the best estimator in such situations.

- However, by suitably transforming the variables, a linear relationship can be obtained between the transformed response variable and the transformed parameters.

Variable Transformation in Regression

- Considering an exponential regression model,

$$\hat{Y} = ae^{bt}$$

- Taking the natural logarithm on both sides, we have

$$\ln\left(\hat{Y}\right) = \ln(a) + bt$$

- The above expression is linear in terms of the parameters and hence, OLS method can be used to estimate the parameters ($\ln(a)$ and b).

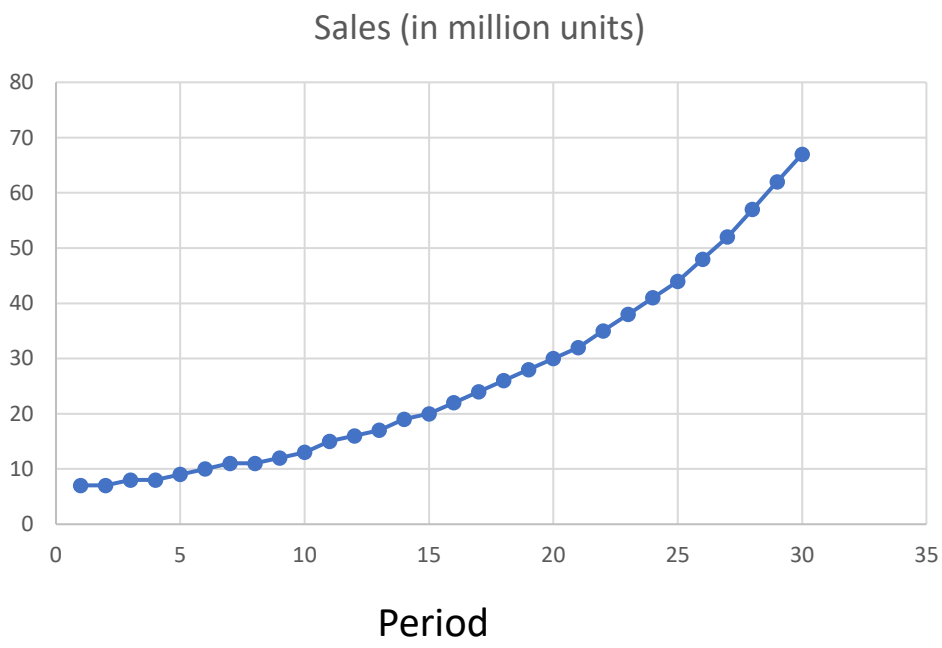
Example

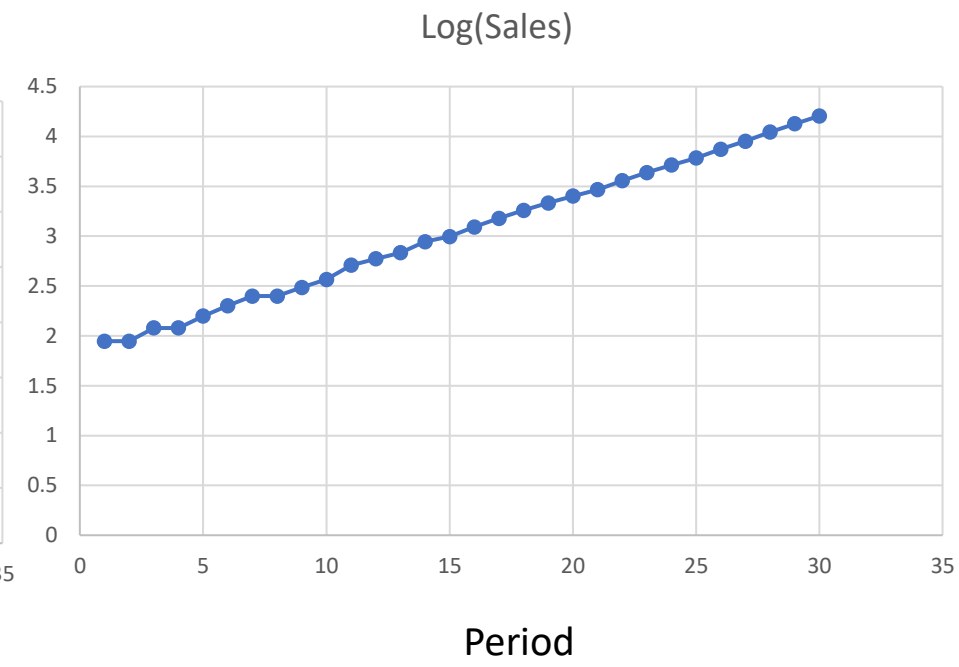
- Consider the following quarterly data for the sales of smartphones. Predict the sales for the 31st quarter.

t	Sales (in million units)	t	Sales (in million units)
1	7	16	22
2	7	17	24
3	8	18	26
4	8	19	28
5	9	20	30
6	10	21	32
7	11	22	35
8	11	23	38
9	12	24	41
10	13	25	44
11	15	26	48
12	16	27	52
13	17	28	57
14	19	29	62
15	20	30	67

Example

- Plotting the sales vs time and the natural logarithm of sales vs time, we have:





Example

- Here, the natural logarithm of the response variable (sales) is linear in terms of the predictor variable (time).

- Hence, an exponential regression model can be used in this scenario.

$$\hat{Y} = ae^{bt}$$

- Taking the natural logarithm on both sides, we have

$$\ln\left(\hat{Y}\right) = \ln(a) + bt$$

Example

- Using OLS method, the parameters $\ln(a)$ and b are estimated as follows:

$$\ln(a) = 1.8087 \;\Rightarrow a = \exp(1.8087) = 6.1024$$
$$b = 0.0796$$

- The regression equation is: $\widehat{Y}_t = ae^{bt}$

$$\boldsymbol{\widehat{Y}_t = 6.1024 * \exp(0.0796t)}$$

- Forecast for the 31st period ($t = 31$) is:

$\widehat{Y}_{31} = 6.1024 * \exp(0.0796 * 31) = \mathbf{71.92\ million\ units}$

Multiple regression

- In multiple regression there is one variable to be forecast and several predictor variables
- The general form of a multiple regression is

$$\hat{Y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \cdots + \beta_n X_n$$

- $\hat{Y}$ is the variable to be forecast
- $X_1, X_2, X_3, \ldots, X_n$ are the predictor variables.
- Each of the predictor variables must be numerical.
- The coefficients (β_0, β_1, β_2 ,…, β_n) measure the effect of each predictor after taking account of the effect of all other predictors in the model.
- The coefficients are calculated using computer packages.

Quadratic regression example

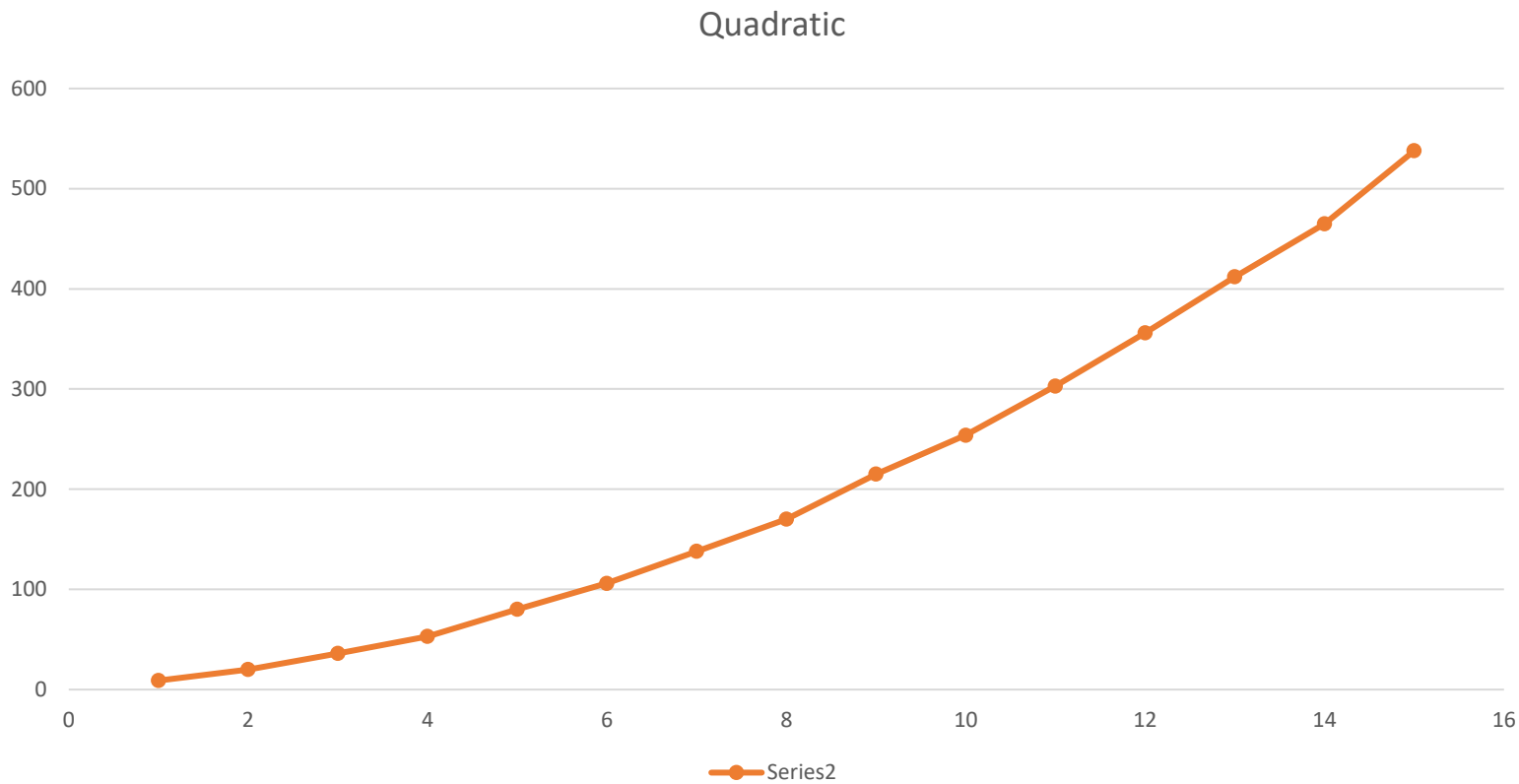
- Consider the following data for the height of a tree measured over the course of its lifetime. Fit a quadratic model and estimate its parameters.

Age of tree	Height of Tree (in)
1	9
2	20
3	36
4	53
5	80
6	106
7	138
8	170
9	215
10	254
11	303
12	356
13	412
14	465
15	538

Quadratic regression example

- The best-fit regression equation is:

$$\hat{Y}_t = 5.623t + 1.998t^2$$



Harmonic Regression (to model cyclic data)

- Considering only the fundamental frequencies,

$$\hat{Y}(t) = A\sin\left(\frac{2\pi}{p}t\right) + B\cos\left(\frac{2\pi}{p}t\right)$$

- Considering first order and second order frequencies,

$$\hat{Y}(t) = A_1\sin\left(\frac{2\pi}{p}t\right) + B_1\cos\left(\frac{2\pi}{p}t\right) + A_2\sin\left(\frac{4\pi}{p}t\right) + B_2\cos\left(\frac{4\pi}{p}t\right)$$

- Considering higher order (up to order n) frequencies,

$$\hat{Y}(t) = \sum_{i=1}^{n} A_i\sin\left(\frac{2\pi i}{p}t\right) + \sum_{i=1}^{n} B_i\cos\left(\frac{2\pi i}{p}t\right)$$

Harmonic Regression (example)

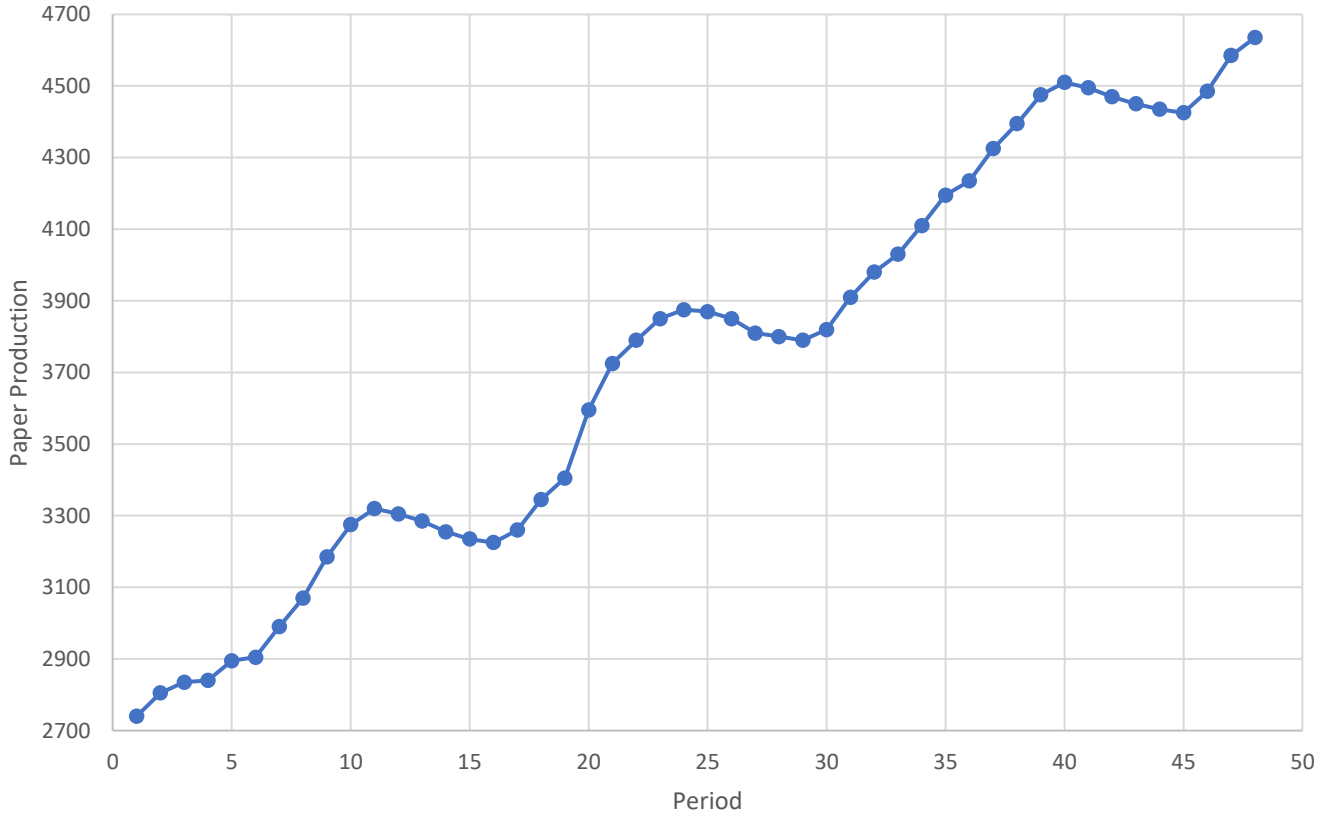
- Consider the following monthly data for paper production in the US over a 4-year period:

Period	Paper_Prod
Jan-17	2740
Feb-17	2805
Mar-17	2835
Apr-17	2840
May-17	2895
Jun-17	2905
Jul-17	2990
Aug-17	3070
Sep-17	3185
Oct-17	3275
Nov-17	3320
Dec-17	3305
Jan-18	3285
Feb-18	3255
Mar-18	3235
Apr-18	3225
May-18	3260
Jun-18	3345
Jul-18	3405
Aug-18	3595
Sep-18	3725
Oct-18	3790
Nov-18	3850
Dec-18	3875

Period	Paper_Prod
Jan-19	3870
Feb-19	3850
Mar-19	3810
Apr-19	3800
May-19	3790
Jun-19	3820
Jul-19	3910
Aug-19	3980
Sep-19	4030
Oct-19	4110
Nov-19	4195
Dec-19	4235
Jan-20	4325
Feb-20	4395
Mar-20	4475
Apr-20	4510
May-20	4495
Jun-20	4470
Jul-20	4450
Aug-20	4435
Sep-20	4425
Oct-20	4485
Nov-20	4585
Dec-20	4635

Harmonic Regression (example)

- From a plot of the production data, it can be clearly seen that the data exhibits cyclicity as well as an increasing trend.



Harmonic Regression (example)

- Assuming an additive model and considering only the first and second order frequencies, the regression equation can be written as:

$$\hat{Y}(t) = A + Bt + C\sin\left(\frac{2\pi}{p}t\right) + D\cos\left(\frac{2\pi}{p}t\right) + E\sin\left(\frac{4\pi}{p}t\right) + F\cos\left(\frac{4\pi}{p}t\right)$$

- Here, the period p is 12 months.

- Considering the sine and cosine functions as independent predictor variables, the model can be treated as a multiple linear regression model with five predictors and six coefficients.

Harmonic Regression (example)

- The estimated value of the coefficients based on method of least squares are:

$$\hat{A} = 2742.29;\ \ \hat{B} = 40.37;\ \ \hat{C} = -8.39;$$
$$\hat{D} = 68.36;\ \ \hat{E} = -4.61;\ \ \hat{F} = -5.26$$

- The regression equation is:

$$\hat{Y}(t) = 2742.29 + 40.37t - 8.39\sin\left(\frac{2\pi}{12}t\right) + 68.36\cos\left(\frac{2\pi}{12}t\right) - 4.61\sin\left(\frac{4\pi}{12}t\right) - 5.26\cos\left(\frac{4\pi}{12}t\right)$$

- The forecast for month 49 (Jan 2021) is:

$$\hat{Y}(49) = 2742.29 + 40.37 * 49 - 8.39\sin\left(\frac{2\pi}{12}49\right) + 68.36\cos\left(\frac{2\pi}{12}49\right) - 4.61\sin\left(\frac{4\pi}{12}49\right) - 5.26\cos\left(\frac{4\pi}{12}49\right)$$

$$\hat{Y}(49) = 4768.4 \text{ units}$$

Harmonic Regression (Seasonality)

- In addition to cyclicity, sine and cosine functions may also be used to model seasonality.

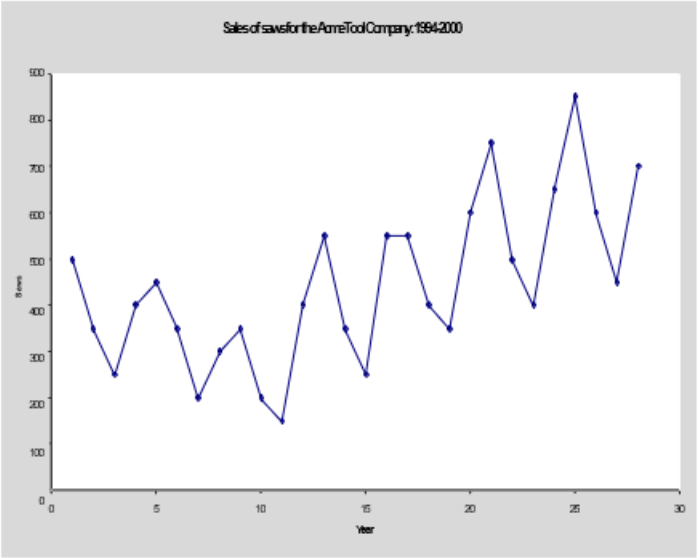
- In cycles, patterns of rise and fall may be repeated over irregular periods of time, and hence, they may require sine and cosine functions of multiple periodicities in order to model them.

- However, seasons always correspond to fixed, known periods of time and hence, they can be modelled using sine and cosine functions of a single periodicity.

Harmonic Regression (Seasonality)

- Considering the same example as in Holt-Winters ES, the quarterly sales data for a company is tabulated below:

Year	Quarter	sales
2014	1	500
	2	350
	3	250
	4	400
2015	1	450
	2	350
	3	200
	4	300
2016	1	350
	2	200
	3	150
	4	400
2017	1	550
	2	350
	3	250
	4	550
2018	1	550
	2	400
	3	350
	4	600
2019	1	750
	2	500
	3	400
	4	650
2020	1	850
	2	600
	3	450
	4	700



Harmonic Regression (Seasonality)

- Since the data contains seasonality and trend, the regression equation can be written as:

$$\hat{Y}(t) = A + Bt + C\sin\left(\frac{2\pi}{p}t\right) + D\cos\left(\frac{2\pi}{p}t\right)$$

- Here the period p is 4.

- The estimated regression coefficients are:

$$\hat{A} = 244.98;\ \ \hat{B} = 13.65;\ \ \hat{C} = 152.93;\ \ \ \hat{D} = 47.07;$$

- The results are tabulated in the following slide:

Year	Quarter	sales	t	sin (pi/2 t)	cos (pi/2 t)	Pred
2014	1	500	1	1.0000	0.0000	411.56
	2	350	2	0.0000	-1.0000	225.21
	3	250	3	-1.0000	0.0000	132.99
	4	400	4	0.0000	1.0000	346.64
2015	1	450	5	1.0000	0.0000	466.15
	2	350	6	0.0000	-1.0000	279.79
	3	200	7	-1.0000	0.0000	187.58
	4	300	8	0.0000	1.0000	401.22
2016	1	350	9	1.0000	0.0000	520.73
	2	200	10	0.0000	-1.0000	334.38
	3	150	11	-1.0000	0.0000	242.16
	4	400	12	0.0000	1.0000	455.81
2017	1	550	13	1.0000	0.0000	575.32
	2	350	14	0.0000	-1.0000	388.97
	3	250	15	-1.0000	0.0000	296.75
	4	550	16	0.0000	1.0000	510.39
2018	1	550	17	1.0000	0.0000	629.91
	2	400	18	0.0000	-1.0000	443.55
	3	350	19	-1.0000	0.0000	351.33
	4	600	20	0.0000	1.0000	564.98
2019	1	750	21	1.0000	0.0000	684.49
	2	500	22	0.0000	-1.0000	498.14
	3	400	23	-1.0000	0.0000	405.92
	4	650	24	0.0000	1.0000	619.57
2020	1	850	25	1.0000	0.0000	739.08
	2	600	26	0.0000	-1.0000	552.72
	3	450	27	-1.0000	0.0000	460.51
	4	700	28	0.0000	1.0000	674.15

Assumptions of Linear Regression

- **Linearity -** Linear relationship exists between the outcome variable and each independent variable. Scatterplots can show whether there is a linear or curvilinear relationship.

- **Normality -** Linear regression assumes that the residuals (errors) are normally distributed.

- **Independence of Errors -** Linear regression assumes that the prediction errors are independent of each other.

- **Homoscedasticity -** Error variance is the same across observations and does not depend on the independent variable(s).

Heteroscedasticity

- Heteroscedasticity is considered to be present in a regression model if the prediction errors from the model appear to have a non-uniform variance.

- It is the absence of homoscedasticity.

- It can lead to inaccurate estimates of the standard errors of regression coefficients and may invalidate statistical tests of significance that require the assumption of equal variance.

Detection of Heteroscedasticity

- The simplest way to detect heteroscedasticity is with the help of a residual plot.
- A residual plot plots the residuals against the predicted values.
- If heteroscedasticity is absent, the residuals will be randomly distributed within a narrow, parallel band centred around the mean (0) and there will be no pattern among the errors.

Detection of Heteroscedasticity

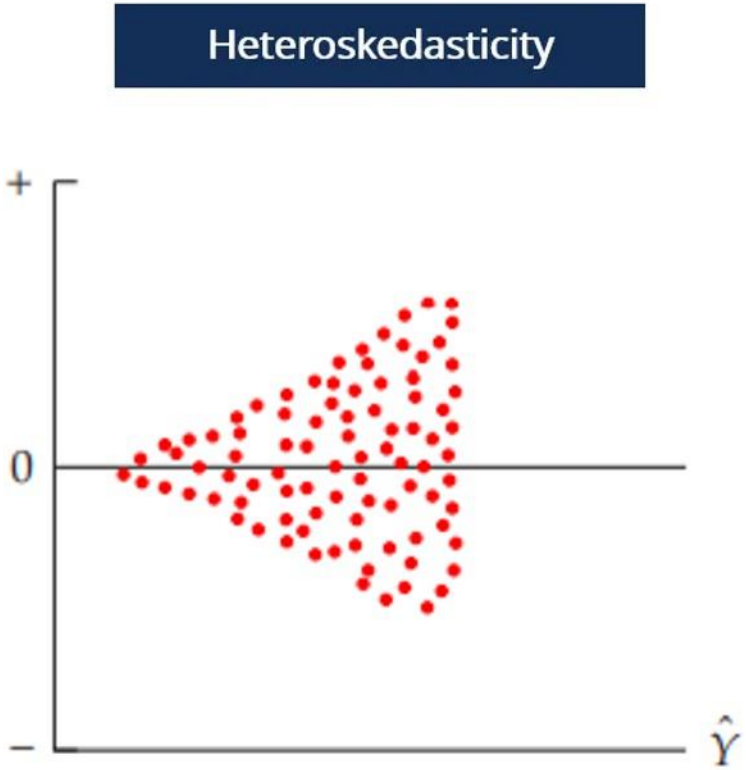
- If heteroscedasticity is present, there will be a consistent pattern in the residual plot.

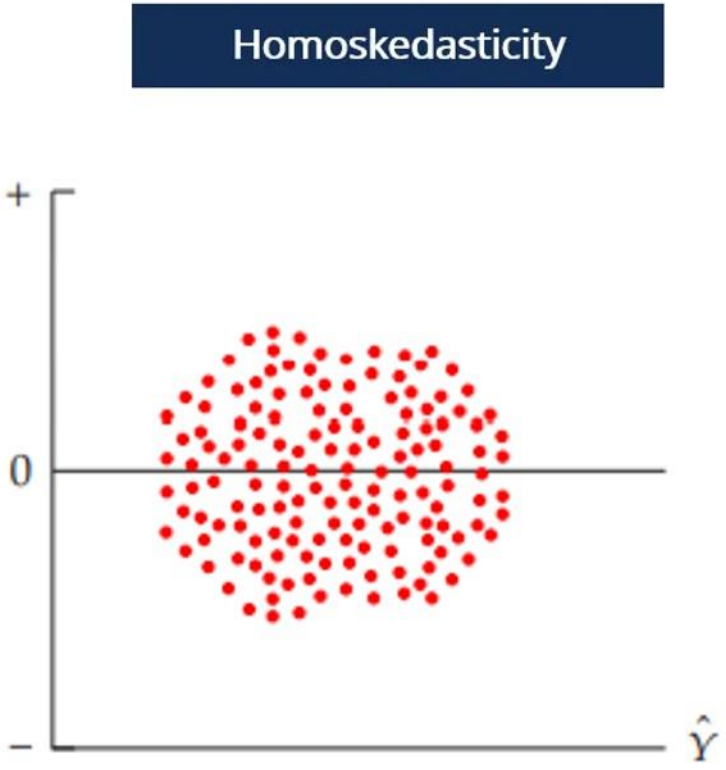
- Additionally, in the case of multiple regression, separate plots of residuals against each independent variable may be constructed to identify the ones that violate homoscedasticity.

- The more rigorous approach to detect heteroscedasticity involves using statistical tests like Breusch-Pagan Test, White Test and Levine Test, which require certain assumptions to be satisfied.

Heteroscedasticity and Homoscedasticity

- Residual Plots for a heteroscedastic and homoscedastic data set respectively





Remedies for Heteroscedasticity

- **Variable Transformation**: By applying suitable transformations to the offending variable(s), the error variance can be made constant.

- **Remediation of other assumption violations:** Sometimes, violation of the assumptions of linearity and normality may lead to heteroscedasticity in the model. By remedying those violations, the effect of heteroscedasticity may disappear.

- **Weighted Least Squares**: Here, each observation is “weighted” by an estimate of its variance and then OLS method is applied on the weighted data to estimate the parameters.

Remedies for Heteroscedasticity

- **Heteroscedasticity-Consistent Standard Errors(HCSE)**:
  - Also known as robust errors, HCSEs are estimates of standard errors of regression coefficients that have been adjusted for heteroscedasticity.
  - Unlike the first two methods, HCSEs don't alter the regression coefficients and are much simpler to use.
  - Four different estimates of HCSE are available, of which one is expressed below: (For a simple regression $\hat{Y} = a + bX$ with n observations and the ith error term being $\hat{u}_i$,)

$$HCSE_1 = \sqrt{\frac{1}{n} \cdot \frac{\frac{1}{n-2}\sum_{i=1}^{n}(X_i - \overline{X})^2 \hat{u}_i^2}{\left[\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2\right]^2}}$$

Example

- The heights and weights of a sample of 100 athletes in a sports academy are tabulated below. Fit a linear regression model to predict the weight of an athlete from his/her height and check for homoscedasticity assumption using graphical inspection.

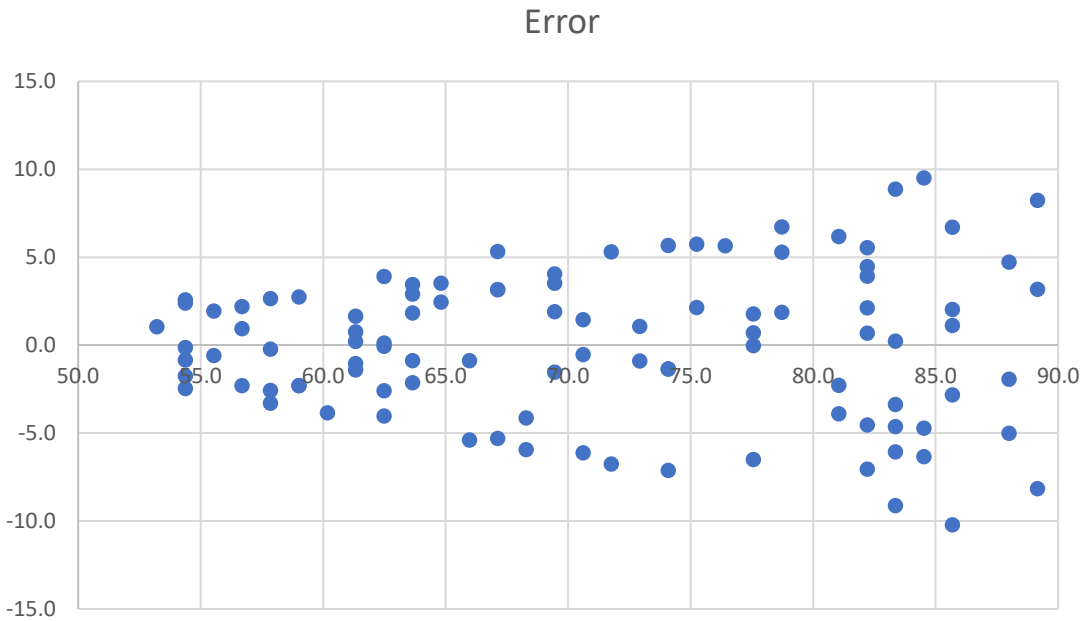
Height (in m)	Weight (kg)	Height (in m)	Weight (kg)	Height (in m)	Weight (kg)	Height (in m)	Weight (kg)
1.73	71.3	1.69	68.3	1.62	58.9	1.87	92.4
1.85	80.0	1.70	65.1	1.81	85.4	1.83	87.2
1.80	77.5	1.73	67.9	1.66	59.9	1.84	77.7
1.87	87.7	1.74	70.1	1.84	75.1	1.73	73.5
1.89	86.1	1.71	61.8	1.60	54.2	1.66	62.1
1.60	51.9	1.68	65.5	1.89	92.7	1.60	53.5
1.73	73.0	1.70	60.6	1.80	78.3	1.66	60.3
1.68	61.5	1.86	79.8	1.81	80.6	1.80	71.1
1.71	72.4	1.85	92.2	1.59	54.3	1.89	83.0
1.69	67.3	1.66	61.5	1.71	70.3	1.85	78.7
1.84	87.7	1.84	84.3	1.87	82.8	1.77	79.7
1.63	55.3	1.74	64.5	1.77	72.7	1.87	75.5
1.72	64.2	1.78	77.4	1.85	74.2	1.74	72.0
1.71	70.3	1.68	67.1	1.60	52.6	1.84	82.9
1.67	62.4	1.64	56.7	1.75	65.0	1.67	66.4
1.60	56.8	1.66	63.0	1.87	86.8	1.61	54.9
1.75	77.1	1.85	83.6	1.86	78.2	1.77	67.0
1.79	82.1	1.63	57.6	1.62	57.6	1.63	54.6
1.90	97.4	1.61	57.5	1.80	79.3	1.63	60.5
1.64	61.7	1.76	72.0	1.67	59.9	1.86	94.0
1.68	62.8	1.83	78.8	1.83	77.1	1.77	72.7
1.81	84.0	1.68	66.5	1.85	77.3	1.90	92.3
1.60	56.9	1.84	86.7	1.90	81.0	1.67	62.6
1.84	86.1	1.64	56.7	1.76	74.0	1.72	62.4
1.78	81.0	1.67	58.5	1.65	56.3	1.62	54.4

Solution

- The regression equation is determined as:

$$\widehat{Weight} = -131.18 + 115.97 * Height$$

- Plotting the residuals against the predicted values, we find that the error variance is not constant, but appears to be increasing proportionally with the predicted value (and hence with the independent variable). Thus, the data set is found to be heteroscedastic by graphical inspection.



Variable Selection in Regression

- In most instances of multiple regression, the researcher has a number of possible independent variables from which to choose for inclusion in the regression equation.

- Sometimes the set of independent variables is exactly specified and the regression model is essentially used in a confirmatory approach.

- But in most instances, the researcher may choose to specify the variables to be included in the model (by explicit specification or using the combinatorial approach) or use the estimation technique to pick and choose among the set of independent variables with either **sequential search** or **constrained processes**.

Sequential Search Methods

- Sequential search methods are general approach of estimating the regression equation.
- A set of variables is defined followed by adding or deleting among these variables until some overall criterion measure is achieved.

- Two types of sequential search approaches are:

(1) stepwise estimation

(2) forward addition and backward elimination.

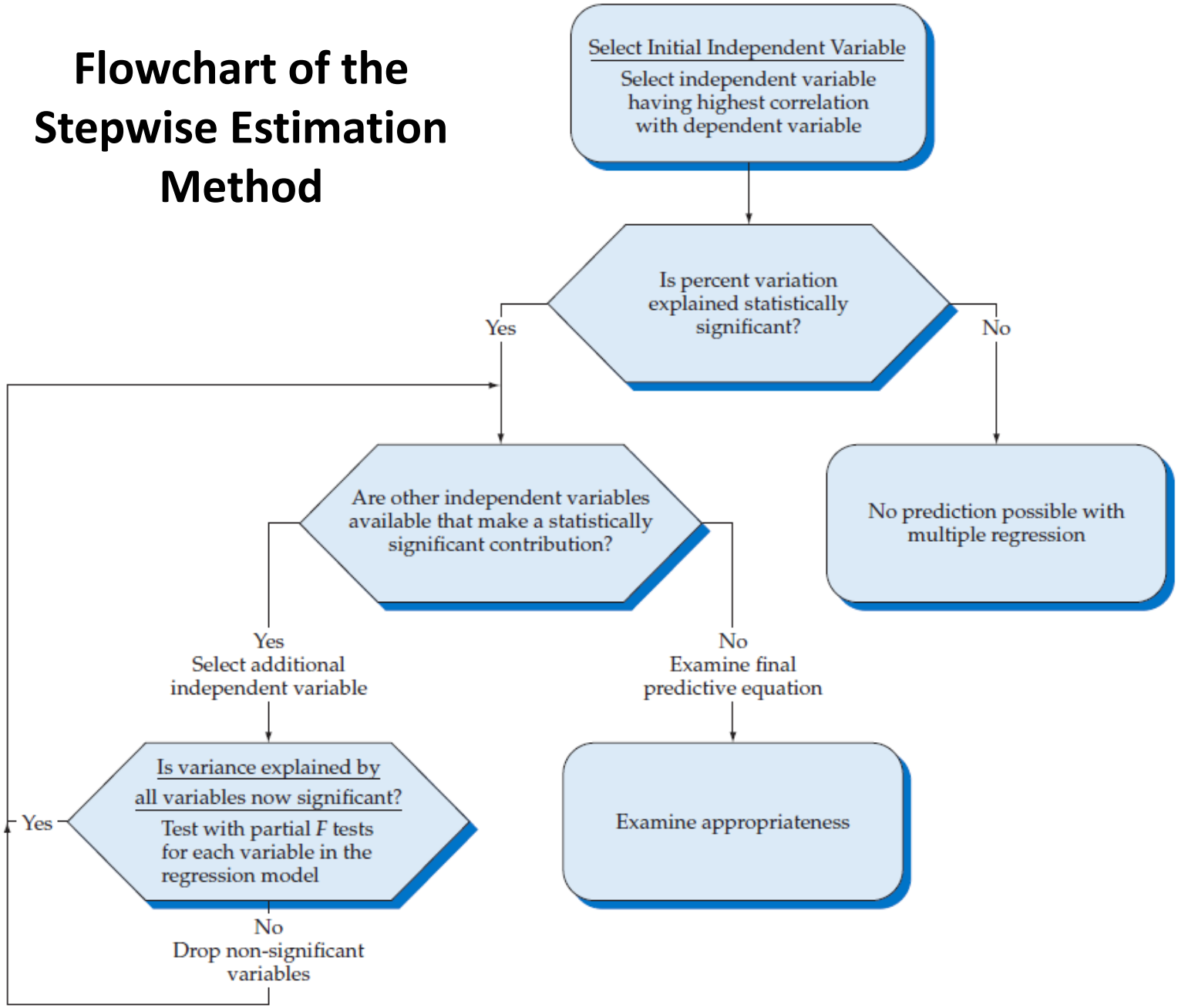
Stepwise Estimation

- It enables the examination of each independent variable's contribution to the regression model.

- The independent variable with the greatest contribution is added first.

- Independent variables are then selected for inclusion based on their incremental contribution over the variable(s) already in the equation.

Flowchart of the Stepwise Estimation Method



Forward Addition And Backward Elimination

- These are largely trial-and-error processes for finding the best regression estimates.
- The forward addition model is similar to the stepwise procedure in that it builds the regression equation starting with a single independent variable.
- The backward elimination procedure starts with a regression equation including all the independent variables and then deletes independent variables that do not contribute significantly.

The primary distinction of the stepwise approach from the forward addition and backward elimination procedures is its ability to add or delete variables at each stage.

- Once a variable is added or deleted in the forward addition or backward elimination schemes, the action cannot be reversed at a later stage.
- Thus, the ability of the stepwise method to add and delete makes it the generally the more preferred method among forecasters.

Forward Addition Steps

- In the forward addition procedure for variable selection, we first perform simple linear regression individually for every predictor-response pair.

- Then, we select the predictor with the highest significance (least p-value, provided it is less than the specified significance level) for entering into the model.

- After the selected predictor is entered into the model, we perform multiple linear regression to predict the response using every combination of the selected predictor with the remaining predictors to determine the next variable to enter the model.

- This process is repeated until no predictor is significant enough to enter the model for the next stage.

Backward Elimination Steps

- In the backward elimination procedure for variable selection, we first perform multiple linear regression considering all predictors at once.

- Then, we select the predictor with the least significance (highest p-value, provided it is greater than or equal to the specified significance level) for leaving the model.

- After the selected predictor is removed from the model, we perform multiple linear regression considering the remaining predictors to determine the next variable to leave the model.

- This process is repeated until no predictor is insignificant enough to leave the model at the current stage.

Example

- An automotive magazine aims to study the effect of different factors on the mileage of a car.

- Preliminary analysis led them towards the conclusion that three factors - engine displacement, engine power and the time it takes to accelerate from 0 kmph to 100 kmph – were especially important in influencing a car's mileage.

- For their experiment, they considered 38 different models of cars (all produced in the year 2020) and measured these four variables for each model. These values are tabulated in the next slide.

- Explore the different variable selection techniques (consider a significance level of $\boldsymbol{\alpha = 0.10}$ for a variable to enter/leave the model) and fit a suitable regression model to predict mileage using the input variables.

Example

Mileage (kmpl)	Engine displacement (litres)	Engine Power (kW)	Acceleration (0-100 kmph) (sec)
22.1	5.7	115.6	14.9
20.3	5.8	105.9	14.3
25.2	4.4	93.2	15
24.2	5.9	111.9	13
39.3	1.6	50.7	16.5
36	2.2	70.8	14.2
35.6	2	72.3	14.7
40.5	1.7	55.9	14.5
26.6	2.1	76.8	15.9
22.3	2.7	93.2	13.6
28.3	2	85.8	15.7
21.2	2.7	99.2	15.8
27	3.8	78.3	15.8
27.3	3.3	63.4	16.7
24.4	3.7	82	18.7
23.7	4.2	89.5	15.1
22.3	5	96.9	15.4
23.1	4.9	96.2	13.4
21.6	5.8	102.9	13.2

Mileage (kmpl)	Engine displacement (litres)	Engine Power (kW)	Acceleration (0-100 kmph) (sec)
23.9	5.2	100.7	15.2
34.7	2.3	65.6	14.4
28.7	2.8	81.3	16.6
44.7	1.4	48.5	15.2
46	1.6	59.7	14.4
35.9	2	59.7	15
41.3	1.5	52.9	14.9
38.7	1.6	50.7	16.6
37.2	2.5	67.1	16
37.7	2.8	85.8	11.3
35.1	2.8	85.8	12.9
43.9	2.5	67.1	13.2
44.8	1.7	52.2	13.2
41.7	1.4	48.5	19.2
48.9	1.5	51.5	14.7
40	1.6	58.2	14.1
28.8	2.4	72.3	14.5
28.2	2	82	12.8
41.8	1.5	52.9	14

Forward Addition – Step 1

- Mileage vs. Displacement

	Coefficients	Standard Error	t Stat	*p*-value
Intercept	45.963	1.966	23.379	2.35E-23
Engine displacement (litres)	-4.643	0.606	-7.666	**4.48E-09**

- Mileage vs. Power

	Coefficients	Standard Error	t Stat	*p*-value
Intercept	61.219	2.783	21.996	1.83E-22
Engine Power (kW)	-0.379	0.036	-10.673	**1.06E-12**

- Mileage vs. Acceleration

	Coefficients	Standard Error	t Stat	*p*-value
Intercept	36.990	13.662	2.708	0.010
Acceleration (0-100 kmph) (sec)	-0.306	0.915	-0.334	**0.740**

- Since Engine Power is significant (p-value $< \alpha = 0.1$) and has the lowest p-value, it is selected for entry into the model.

Forward Addition – Step 2

- Mileage vs. Power and Displacement

	Coefficients	Standard Error	t Stat	*p*-value
Intercept	59.876	3.412	17.546	6.38E-19
Engine Power (kW)	-0.335	0.073	-4.595	5.41E-05
Engine displacement (litres)	-0.683	0.989	-0.691	**0.494**

- Mileage vs. Power and Acceleration

	Coefficients	Standard Error	t Stat	*p*-value
Intercept	87.739	6.674	13.146	4.22E-15
Engine Power (kW)	-0.412	0.030	-13.588	1.60E-15
Acceleration (0-100 kmph) (sec)	-1.620	0.383	-4.232	**1.59E-04**

- Since Acceleration is significant and has the lowest p-value, it is selected as the second variable to enter into the model.

Forward Addition – Step 3

- Mileage vs. Power, Acceleration and Displacement

	Coefficients	Standard Error	t Stat	*p*-value
Intercept	87.350	7.289	11.984	9.35E-14
Engine Power (kW)	-0.404	0.063	-6.422	2.45E-07
Acceleration (0-100 kmph) (sec)	-1.610	0.394	-4.091	2.49E-04
Engine displacement (litres)	-0.120	0.833	-0.144	**0.886**

- Since Displacement is insignificant and it is the only remaining predictor yet to be entered into the model, we find that no further variable entry is possible and the process terminates.

- The final regression equation based on variable selection by forward addition procedure is (based on the coefficients from the last table in Step 2):

$$\widehat{Mileage} = 87.739 - 0.412 * Power - 1.620 * Acceleration$$

Backward Elimination

Step 1:

- Mileage vs. Displacement, Power and Acceleration

	Coefficients	Standard Error	t Stat	*p*-value
Intercept	87.350	7.289	11.984	9.35E-14
Engine displacement (litres)	-0.120	0.833	-0.144	**0.886**
Engine Power (kW)	-0.404	0.063	-6.422	**2.45E-07**
Acceleration (0-100 kmph) (sec)	-1.610	0.394	-4.091	**2.49E-04**

- Since Displacement has the highest p-value and is insignificant, it is the first predictor to be removed from the model.

Step 2:

- Mileage vs. Power and Acceleration

	Coefficients	Standard Error	t Stat	*p*-value
Intercept	87.739	6.674	13.146	4.22E-15
Engine Power (kW)	-0.412	0.030	-13.588	**1.60E-15**
Acceleration (0-100 kmph) (sec)	-1.620	0.383	-4.232	**1.59E-04**

- Here, both variables are significant and no further variable removal is possible. The final regression equation is:

$$\widehat{Mileage} = 87.739 - 0.412 * Power - 1.620 * Acceleration$$

Stepwise Selection – Step 1

- Mileage vs. Displacement

	Coefficients	Standard Error	t Stat	*p*-value
Intercept	45.963	1.966	23.379	2.35E-23
Engine displacement (litres)	-4.643	0.606	-7.666	**4.48E-09**

- Mileage vs. Power

	Coefficients	Standard Error	t Stat	*p*-value
Intercept	61.219	2.783	21.996	1.83E-22
Engine Power (kW)	-0.379	0.036	-10.673	**1.06E-12**

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Stepwise Selection – Step 2

- Mileage vs. Power and Displacement

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Intercept	59.876	3.412	17.546	6.38E-19
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	Coefficients	Standard Error	t Stat	*p*-value
Intercept	87.739	6.674	13.146	4.22E-15
Engine Power (kW)	-0.412	0.030	-13.588	**1.60E-15**
Acceleration (0-100 kmph) (sec)	-1.620	0.383	-4.232	**1.59E-04**

- Since Acceleration is significant and has the lowest p-value, it is selected as the second variable to enter into the model.
- Now, in stepwise selection procedure, it is needed to be checked if the entry of this new variable has led to previously entered variables becoming insignificant. Clearly, in this case, the previously entered variable (Power) continues to be significant. So both variables are retained going into Step 3.

Stepwise Selection – Step 3

- Mileage vs. Power, Acceleration and Displacement

	Coefficients	Standard Error	t Stat	*p*-value
Intercept	87.350	7.289	11.984	9.35E-14
Engine Power (kW)	-0.404	0.063	-6.422	**2.45E-07**
Acceleration (0-100 kmph) (sec)	-1.610	0.394	-4.091	**2.49E-04**
Engine displacement (litres)	-0.120	0.833	-0.144	**0.886**

- Since Displacement is insignificant and it is the only remaining predictor yet to be entered into the model, we find that no further variable entry is possible and the process terminates.

- The final regression equation based on variable selection by stepwise selection procedure is (based on the coefficients from the last table in Step 2):

$$\widehat{Mileage} = 87.739 - 0.412 * Power - 1.620 * Acceleration$$

Constrained Methods for Variable Selection

- These methods are also known as regularisation methods.

- There are 2 such methods:
  - Ridge regression
  - LASSO regression

- Both methods add a penalty term (consisting of a function of regression coefficients) to the loss function to be minimized for estimating the regression parameters.

- They have the effect of ‘shrinking’ the regression coefficients and prevent overfitting.

Ridge Regression

- It is also known as L2 regularisation.
- Consider the following multiple linear regression equation:

$$\hat{y} = w_1 x_1 + w_2 x_2 + ... + w_N x_N + b$$

- Under Ridge regression approach, the loss function is added to by a penalty term that is the product of the Ridge parameter ($\lambda > 0$) and the sum of squares of the regression coefficients.

$$Loss = Error(y, \hat{y}) + \lambda \sum_{i=1}^{N} w_i^2$$

- Here, $Error(y, \hat{y})$ is some metric of forecast error for the model, e.g. sum of squared errors.

LASSO Regression

- LASSO stands for Least Absolute Shrinkage and Selection Operator.
- It is also known as L1 regularisation.
- It is similar to the Ridge regression approach, the only difference is that the penalty term includes the sum of absolute values of regression coefficients instead of the sum of squares of the regression coefficients.

$$Loss = Error(y, \hat{y}) + \lambda \sum_{i=1}^{N} |w_i|$$

- It allows certain regression coefficients to be set to zero, thereby completely eliminating insignificant factors from the model.

Multicollinearity and forecasting

Multicollinearity occurs when similar information is provided by two or more of the predictor variables in a multiple regression. It can occur in a number of ways:

1. Two predictors are highly correlated with each other (that is, they have a correlation coefficient close to +1 or -1). In this case, knowing the value of one of the variables tells you a lot about the value of the other variable.

2. A linear combination of predictors is highly correlated with another linear combination of predictors. In this case, knowing the value of the first group of predictors tells you a lot about the value of the second group of predictors. Hence, they are providing similar information.

Measuring Multicollinearity

- The metric commonly used to measure multi-collinearity is the Variance Inflation Factor (VIF).

- VIF is a measure of the degree to which the square of the standard error of a regression coefficient has been increased due to multicollinearity.

- For example, suppose the VIF corresponding to a regression coefficient is 4. This implies that the square of the standard error of that coefficient has been increased by a factor of 4, or the standard error of that coefficient has increased by a factor of $\sqrt{4}$ = 2, due to the effects of multicollinearity.

- This might have led to incorrect inferences of the associated variable's significance in the regression model.

Variance Inflation Factor

- Considering the following MLR model (with n predictors):

$$\hat{Y} = b_0 + b_1 X_1 + b_2 X_2 + \cdots + b_n X_n$$

- The VIF for each predictor k is given by:

$$VIF_k = \frac{1}{1 - R_k^2}$$

- Where R_k^2 is the coefficient of *multiple* determination considering X_k as the dependent variable and all other X_i ($i \in \{1,2,3,\ldots,n\} \backslash \{k\}$) as independent variables

- The amount of VIF_k that represents an acceptable level of multicollinearity is highly problem specific and difficult to generalize.

Variance Inflation Factor

- Consider the same mileage example from before. The regression equation is:

$$\widehat{\boldsymbol{Mileage}} = \mathbf{87.739} - \mathbf{0.412} * \boldsymbol{Power} - \mathbf{1.620} * \boldsymbol{Acceleration}$$

- Since there are only two significant predictors in this case, their respective VIFs will be one and the same.

- Treating $Power$ as the dependent variable and $Acceleration$ as the independent variable, regression is performed and the best fit model is found to be:

$$\widehat{\boldsymbol{Power}} = \mathbf{123.29} - \mathbf{3.192} * \boldsymbol{Acceleration}$$

Variance Inflation Factor

- The coefficient of determination for this regression is found to be:

$$R^2_{Power} = 0.0638$$

- The VIF is given by:

$$VIF_{Power} = \frac{1}{1 - R^2_{Power}} = \frac{1}{1 - 0.0638} = \frac{1}{0.9362} = \mathbf{1.0682}$$

- Here, the VIF is only marginally greater than 1, implying that there is very little correlation between the predictors, and hence, the effects of multicollinearity are not significant in this model.

Adjusted Coefficient of Determination

- **Adjusted $\boldsymbol{R^2}$**: One disadvantage of the coefficient of determination (R^2) is that addition of more predictor variables will inevitably not decrease its value.

- In other words, even if additional non-significant variables are introduced into the regression model, the value of R^2 will either stay constant or increase, thereby encouraging overfitting.

- To overcome this drawback, the adjusted R^2 is calculated. This metric is adjusted for the no. of predictors (k) in the model and it increases only when an added variable improves the model more than what would be expected by chance.

$$R^2{}_{adj} = 1 - \frac{(1 - R^2)(n - 1)}{n - k - 1}$$

Adjusted Coefficient of Determination

- Consider the same mileage example from before (Sample size $n = 38$). The regression equation is:

$$\widehat{\boldsymbol{Mileage}} = \mathbf{87.739} - \mathbf{0.412} * \boldsymbol{Power} - \mathbf{1.620} * \boldsymbol{Acceleration}$$

- The coefficient of multiple determination for this two-predictor model is found to be:

$$R^2 = R^2_{(Pow,Acc)} = 0.8411$$

- The adjusted coefficient of multiple determination for this model is calculated as:

$$R^2_{Adj\ (Pow,Acc)} = 1 - \frac{(1 - R^2)(n - 1)}{n - k - 1} = 1 - \frac{(1 - 0.8411)(38 - 1)}{38 - 2 - 1}$$

$$\boldsymbol{R^2_{Adj\ (Pow,Acc)} = 0.8321}$$

Adjusted Coefficient of Determination

- We had previously established during the variable selection procedure that the third variable (Engine Displacement) is not significant.

- Now, we deliberately include this variable in the model to see its effect on R^2 and R^2_{adj}. The regression equation is:

$$\widehat{Mileage} = 87.350 - 0.404 * Power - 1.610 * Acceleration - 0.120 * Displacement$$

- The coefficient of multiple determination for this three-predictor model is found to be:

$$R^2 = R^2_{Pow,Acc,Disp} = 0.8412$$

Adjusted Coefficient of Determination

- We find that there is a marginal increase in the value of R^2 despite adding an insignificant variable.

- Now, we compute the adjusted coefficient of multiple determination for this three-predictor model.

$$R^2_{Adj\ (Pow,Acc,Disp)} = 1 - \frac{(1 - R^2)(n - 1)}{n - k - 1} = 1 - \frac{(1 - 0.8412)(38 - 1)}{38 - 3 - 1}$$

$$\boldsymbol{R^2_{Adj\ (Pow,Acc,Disp)} = 0.8272}$$

- We find that the adjusted coefficient of multiple determination has ***decreased*** after the addition of an insignificant variable.

Regression vs Linear Programming Model

- Given a dataset with n observations,

x	y
1	17500
2	19000
3	23000
4	33000
5	37250
6	35500
7	41500

Run the regression model on the dataset to get the intercept (a) and the slope (b). Calculate forecasted value $(\hat{y} = a + bx)$ for each observation.

Results with Regression

$$a = 12500$$
$$b = 4258.93$$
$$\hat{y} = 12500 + 4258.93x$$

x	y	$\hat{y}$
1	17500	16758.93
2	19000	21017.86
3	23000	25276.79
4	33000	29535.72
5	37250	33794.65
6	35500	38053.57
7	41500	42312.5

- Mean Absolute Percentage Error (MAPE)= **7.67%**

Linear Programming Model

Let E_i (percentage error), a and b (regression coefficients) be the decision variables.

Objective:

$$Min\ Z = \frac{1}{n}\sum_{i=1}^{n} E_i$$

subject to

1. $\hat{y}_i = a + bx_i;\ \forall i \in \{1,2,3,\dots,n\}$
2. $E_i \geq (\hat{y}_i - y_i)/y_i;\ \forall i \in \{1,2,3,\dots,n\}$
3. $E_i \geq (y_i - \hat{y}_i)/y_i;\ \forall i \in \{1,2,3,\dots,n\}$

$E_i \geq 0;\ \forall i \in \{1,2,3,\dots,n\}$

a, b unrestricted in sign

Results with LP using GAMS:

$$Z = 0.0719$$

$$a = 13500, \qquad b = 4000$$

$$\hat{y} = 13500 + 4000x$$

X	Y	Y^	err	err/Y	13500	a
1	17500	17500	0	0	4000	b
2	19000	21500	2500	0.131579		
3	23000	25500	2500	0.108696		
4	33000	29500	3500	0.106061		
5	37250	33500	3750	0.100671		
6	35500	37500	2000	0.056338		
7	41500	41500	0	0		
			MAPE=	7.190634	%	

MAPE = 7.19%

MAPE obtained using Linear programming model (7.19%) is less than that obtained using regression (7.67%).

LP model considering costs of overestimation and underestimation

- Let C_i (cost), a and b (regression coefficients) be the decision variables.
- **Objective:** $Min\ Z = \sum_{i=1}^{n} C_i$

  subject to

  1. $\hat{y}_i = a + bx_i;\ \forall i \in \{1,2,3,\dots,n\}$
  2. $C_i \geq (\hat{y}_i - y_i)p_o;\ \forall i \in \{1,2,3,\dots,n\}$
  3. $C_i \geq (y_i - \hat{y}_i)p_u;\ \forall i \in \{1,2,3,\dots,n\}$

  $C_i \geq 0; \forall i \in \{1,2,3,\dots,n\}$

  a, b unrestricted in sign
- Here, p_o and p_u are the unit costs of overestimating and underestimating the response variable, respectively.
- Special case: $p_o = p_u = 1$. Here, the objective of minimizing the total cost becomes equivalent to minimizing the mean absolute deviation.

Stationarity in time-series

- Stationary time-series is one whose properties do not depend on the time at which the series is observed.
- It does not matter when you observe it, it should look much the same at any period of time.
- So time-series with trends, or with seasonality, are not stationary – the trend and seasonality will affect the value of the time-series at different times.
- In general, a stationary time-series will have no predictable patterns in the long-term.

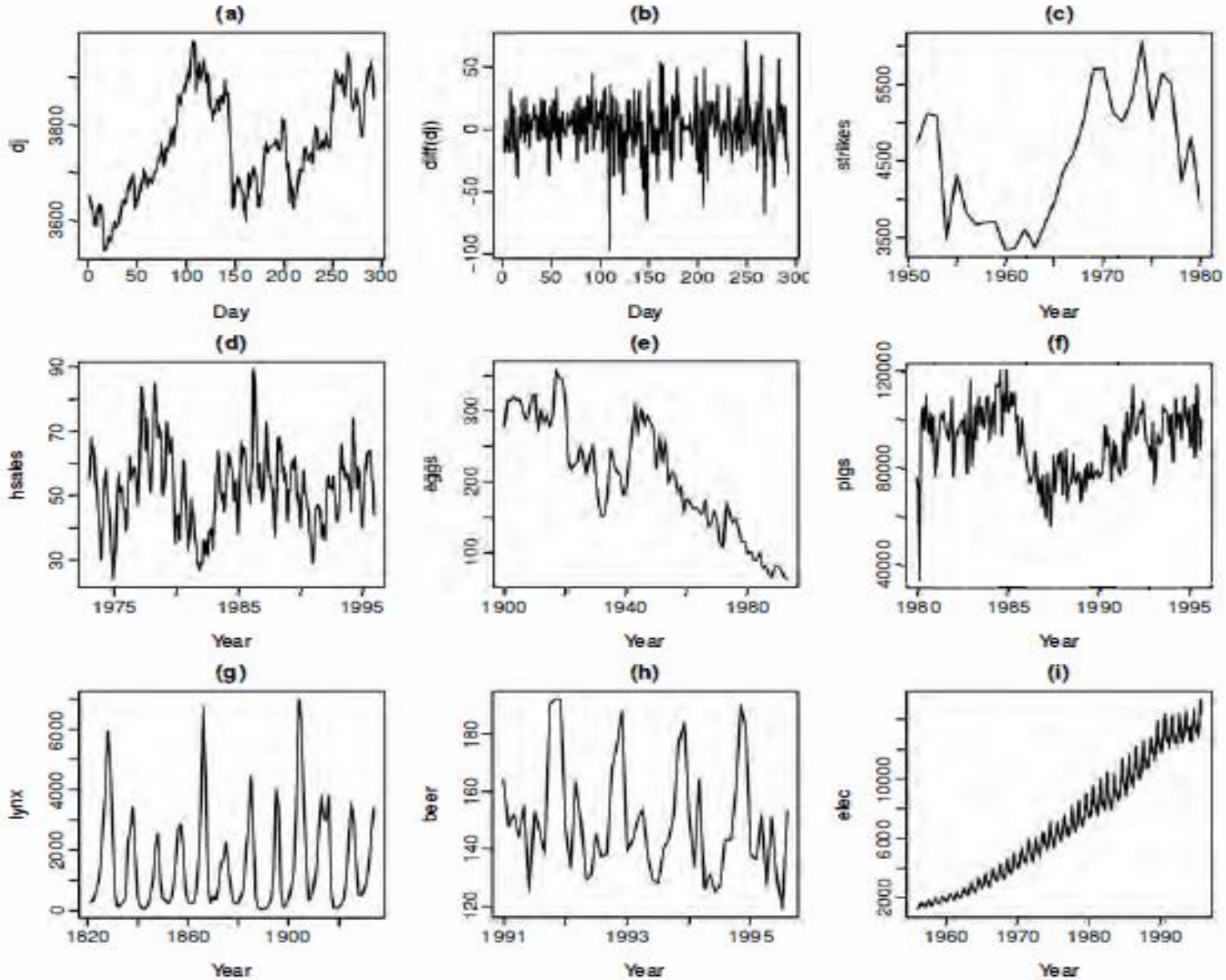
Stationarity in time-series

- In general, a series of observations over time $\{y_t\}$ is stationary if the following conditions hold:

$$\mu_t = \mu;\ \forall t$$
$$\sigma_t = \sigma;\ \forall t$$
$$\rho(y_t, y_{t+h}) = \rho_h$$

- In other words, for a series to be stationary, the mean and variance of the time-series must not change over time and the correlation coefficient of any pair of observations that are h periods apart must be the same, i.e., it should be independent of their actual positions in the time-series.

Which of these are stationary ?



- Obvious seasonality rules out series (d), (h) and (i).
- Trend rules out series (a), (c), (e), (f) and (i).
- Increasing variance also rules out (f).
- That leaves only (b) and (g) as stationary series.

Differencing

- Computing the differences between consecutive observations is known as **differencing.**

- It is one way of converting a non-stationary time-series into a stationary time-series.

- If y_t and y_{t-1} are the observations of the response variable in periods t and $t-1$ respectively, then the differenced observation in period t is given by,

$$y_t' = y_t - y_{t-1}$$

Differencing

- Transformations such as logarithms can help to stabilize the variance of a time-series.
- Differencing can help stabilize the mean of a time-series by removing changes in the level of a time-series.
- It can also help in eliminating (or reducing) the effects of trend and seasonality.

Second-Order Differencing

- In many cases, differencing only once may not be sufficient to make a non-stationary time-series stationary.

- However, in some of those cases, the *differences of those differences* may be stationary. Such differencing is called second-order differencing.

$$y_t'' = y_t' - y_{t-1}' = (y_t - y_{t-1}) - (y_{t-1} - y_{t-2})$$

$$y_t'' = y_t - 2y_{t-1} + y_{t-2}$$

Auto Regressive Models/Process

- In a multiple regression model, we forecast the variable of interest using a linear combination of predictors.

- In an autoregression model, we forecast the variable of interest using a linear combination of *past values of the variable.*

- The term *autoregression* indicates that it is a regression of the variable against itself.

- Thus an autoregressive model of order *p* can be written as

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

- Where, ε_t is white noise.

- This is like a multiple regression but with *lagged values* of y_t as predictors. We refer to this as an **AR(p) model.**

- Autoregressive models are remarkably flexible at handling a wide range of different time-series patterns.

- The variance of the error term ε_t will only change the scale of the series, not the patterns.

AR(1) Process

- Consider the following AR(1) process:

$$AR(1): (X_t - \mu) = \varphi_1 (X_{t-1} - \mu) + \varepsilon_t$$

where, X_t is the observed response in period t,
μ is the mean of the observed responses,
φ_1 is the coefficient of the lagged response,
ε_t is the residual in period t

- For simplicity, let the mean-adjusted observations be represented by $\dot{X}_t = X_t - \mu$

- The AR(1) process can be written as:

$$\dot{X}_t = \varphi_1 \dot{X}_{t-1} + \varepsilon_t$$

AR(1) Process

- The regression equation can be written as:

$$\hat{X}_t = X_t - \varepsilon_t = (\mu + \dot{X}_t) - \varepsilon_t = (\mu + \varphi_1 \dot{X}_{t-1} + \varepsilon_t) - \varepsilon_t$$

$$\boldsymbol{\hat{X}_t = \mu + \varphi_1 \dot{X}_{t-1}}$$

- An AR(1) process aims to predict the response using a single lagged value as a predictor and hence, it is analogous to simple linear regression.

- The OLS estimate of φ_1 is given by,

$$\varphi_1 = \frac{Cov(\dot{X}_t, \dot{X}_{t-1})}{Var(\dot{X}_{t-1})}$$

Example

- Consider the following demand for TV sets over a 12-month period. Using AR(1) process, predict the demand for the 13th month. Also, compare the AR(1) model with MA(1) model and a five-period simple moving average using any error metric.

Period	Demand X_t
1	106
2	110
3	118
4	105
5	115
6	100
7	112
8	106
9	118
10	102
11	112
12	110

AR(1) Process

Period	Demand X_t	$\dot{X}_t$	$\dot{X}_{t-1}$	$\dot{X}_t\dot{X}_{t-1}$	$(\dot{X}_t)^2$
1	106	-3.5			12.25
2	110	0.5	-3.5	-1.75	0.25
3	118	8.5	0.5	4.25	72.25
4	105	-4.5	8.5	-38.25	20.25
5	115	5.5	-4.5	-24.75	30.25
6	100	-9.5	5.5	-52.25	90.25
7	112	2.5	-9.5	-23.75	6.25
8	106	-3.5	2.5	-8.75	12.25
9	118	8.5	-3.5	-29.75	72.25
10	102	-7.5	8.5	-63.75	56.25
11	112	2.5	-7.5	-18.75	6.25
12	110	0.5	2.5	1.25	0.25
Sum	**1314**			**-256.25**	**379**
Average	**109.5**			**-23.3**	**31.6**

AR(1) Process

- The data from the previous table can be used to estimate the value of φ_1 for the AR Process.

$$\varphi_1 = \frac{Cov(\dot{X}_t, \dot{X}_{t-1})}{Var(\dot{X}_{t-1})}$$

- Since every $\dot{X}_{t-1}$ is nothing but the lagged value of the corresponding $\dot{X}_t$, their variances are equal. Hence, $Var(\dot{X}_{t-1})$ can be replaced by $Var(\dot{X}_t)$

$$\varphi_1 = \frac{Cov(\dot{X}_t, \dot{X}_{t-1})}{Var(\dot{X}_t)} = \frac{E(\dot{X}_t \dot{X}_{t-1})}{E(\dot{X}_t^2)} = \frac{-23.3}{31.6} = \mathbf{-0.7}$$

AR(1) Process

- The regression model can then be written as:

$$\hat{X}_t = \mu + \varphi_1 \dot{X}_{t-1}$$

$$\hat{X}_t = 109.5 - 0.7\dot{X}_{t-1}$$

- For period $t = 13$, the forecasted value is:

$$\hat{X}_{13} = 109.5 - 0.7\dot{X}_{12}$$

$$\hat{X}_{13} = 109.5 - 0.7 * 0.5$$

$$\boldsymbol{\hat{X}_{13} = 109.2}$$

AR(1) Process

- Residual sum of squares from forecasts using AR(1) model:

Period	Demand X_t	Forecast $\widehat{X}_t$	Error ε_t	ε_t^2
1	106			
2	110	112	-2	4
3	118	109.2	8.8	77.44
4	105	103.6	1.4	1.96
5	115	112.7	2.3	5.29
6	100	105.7	-5.7	32.49
7	112	116.2	-4.2	17.64
8	106	107.8	-1.8	3.24
9	118	112.2	5.8	33.64
10	102	103.6	-1.6	2.56
11	112	115	-3	9
12	110	109.2	0.8	0.64
			Sum	187.9
			Average	17.08

Simple 5-Period Moving Average(MA) Model

Period	Demand X_t	Forecast $\widehat{X}_t$	Error ε_t	ε_t^2
1	106			
2	110			
3	118			
4	105			
5	115			
6	100	110.8	-10.8	116.6
7	112	109.6	2.4	5.76
8	106	110	-4	16
9	118	107.6	10.4	108.2
10	102	110.2	-8.2	67.24
11	112	107.6	4.4	19.36
12	110	110	0	0
Sum	**1314**			**333.2**
Average	**109.5**			**47.6**

Moving Average Model

- A Moving Average (MA) model is similar to an AR model, but uses lagged residuals instead of lagged observations to predict the response.

- An MA model of order q is expressed as:

$$MA(q)\colon X_t - \mu = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots + \theta_q \varepsilon_{t-q}$$

- The most simple type of MA model (MA(1)) is given by:

$$MA(1)\colon X_t - \mu = \varepsilon_t + \theta_1 \varepsilon_{t-1}$$

Moving Average Model

- The corresponding regression equation is:

$$\widehat{X}_t = \mu + \theta_1 \varepsilon_{t-1}$$

- Estimating MA parameters is more difficult than estimating AR parameters, as the lagged error terms are not observable.

- Iterative non-linear fitting procedures need to be used instead of OLS for estimating the MA parameters.

Moving Average Model

- For the same example, an MA(1) model with $\theta_1 = -0.5$ is fit.
- Here $\mu = 109.5$, therefore $\hat{X}_t = 109.5 - 0.5 * \varepsilon_{t-1}$

Period	Demand X_t	Forecast $\hat{X}_t$	Error ε_t	ε_t^2
1	106			
2	110	111.3	-1.3	1.7
3	118	110.2	7.8	60.8
4	105	105.6	-0.6	0.4
5	115	109.8	5.2	27.0
6	100	106.9	-6.9	47.6
7	112	113	-1	1
8	106	110	-4	16
9	118	111.5	6.5	42.3
10	102	106.3	-4.3	18.5
11	112	111.7	0.3	0.1
12	110	109.4	0.6	0.4
Sum	1314			215.7
Average	109.5			19.6

ARMA(p, q) Models

- An ARMA model combines features of AR models and MA models.

- It is defined by two parameters p and q, where p is the order of the AR part of the model and q is the order of the MA part of the model.

- In other words, an ARMA(p, q) model will use p lagged observations and q lagged residuals to predict the response.

- A little consideration will show that an AR(p) process and an MA(q) process are special cases of an ARMA(p, q) process obtained by setting $q = 0$ and $p = 0$, respectively.

ARMA (p, q) Models

- ARMA(1, 1):

$$(X_t - \mu) = \phi_1 (X_{(t-1)} - \mu) + \varepsilon_t + \theta_1 \varepsilon_{(t-1)}$$

- ARMA(2, 1):

$$(X_t - \mu) = \phi_1 (X_{(t-1)} - \mu) + \phi_2 (X_{(t-2)} - \mu) + \varepsilon_t + \theta_1 \varepsilon_{(t-1)}$$

- ARMA(1, 2):

$$(X_t - \mu) = \phi_1 (X_{(t-1)} - \mu) + \varepsilon_t + \theta_1 \varepsilon_{(t-1)} + \theta_2 \varepsilon_{(t-2)}$$

ARIMA($\boldsymbol{p}, \boldsymbol{d}, \boldsymbol{q}$) Models

- ARIMA(p, d, q) models are extensions of ARMA(p, q) models that incorporate differencing, thereby allowing non-stationary time-series to be modelled.

- The ‘I’ in ARIMA stands for ‘Integrated’, which represents the use of differencing of observations to make the time-series stationary.

- In addition to using p lagged observations and q lagged residuals, an ARIMA model includes another parameter d, which denotes the degree of differencing (the number of times the raw observations have been differenced).

ARIMA(p, d, q) Models

- These models have become very popular in forecasting literature over recent years because of their high versatility, flexibility and ability to handle complex time-series that cannot be easily modelled by other approaches.

- Virtually any other forecasting model can be described as a special case of an ARIMA(p, d, q) model.

- Sophisticated ARIMA implementations are included with most modern day statistical software packages.

Box-Jenkins Modelling

- The Box-Jenkins method is a systematic approach to modelling ARIMA processes.
- It was introduced by statisticians George Box and Gwilym Jenkins in 1970 and greatly popularized through a book that they authored.
- It involves identifying a suitable ARIMA model for a time-series data by analysing its features, fitting that model to the data and using it to make forecasts.

Box-Jenkins Modelling

- The Box-Jenkins approach can be summarized in 5 steps:

  1. Data preparation
  2. Model selection
  3. Parameter estimation
  4. Model Checking
  5. Forecasting

Box-Jenkins Modelling

- **Data Preparation:**
  - This involves cleaning the data and performing necessary transformations and differencing to stabilize the variance of the data and minimize the effects of obvious patterns like trend and seasonality.

- **Model Selection:**
  - The model selection may be done using methods such as Autocorrelation function (ACF), Partial Autocorrelation function (PACF) and Extended Autocorrelation function (EACF) methods. Recently, more advanced selection techniques like Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) have become more prominent.

Box-Jenkins Modelling

- **Estimation:**

- Once the model to be used has been identified (i.e., the values of p and q), we need to estimate the parameters φ_i $(i = 1,2,3, \ldots, p)$ and θ_i $(i = 1,2,3, \ldots, q)$.

- The *maximum likelihood estimation* (MLE) technique finds the values of the parameters which maximize the probability of obtaining the data that we have observed.

- For ARIMA models, MLE is very similar to the *least squares* estimates that would be obtained by minimizing

$$\sum_{t=1}^{T} e_t^2$$

Box-Jenkins Modelling

- **Model Checking:**
  - This involves validating the model assumptions and verifying that the model is indeed adequate for its intended purpose. If any inadequacies are found, we return to Step 2 and try to fit a better model.

- **Forecasting:**
  - Once the model has been deemed fit for purpose, the final step is to use the model to make forecasts. Feedback from the performance of the model may be used to further to make minor modifications to it and refine it.

Backcasting

- In some cases, it may be highly useful to make predictions in reverse time, i.e., use present information to make predictions of the past.

- This process is called Backcasting and it is still a relatively less-known concept.

- Backcasting may have interesting applications in fields such geology, history, archaeology and astrophysics, where unknowable happenings of the past may be estimated (backcasted) using information available in the present.

Neural Network Models in Forecasting

- Neural Networks (NNs) are mathematical models inspired by the functioning of biological neurons.

- NNs are built from simple processing elements (PEs) called nodes, units, or neurons, and contain certain adjustable parameters ("weights").

- NNs use rules to modify these parameters upon the presentation of data, specifically to "learn from the environment”.

Neural Network Models in Forecasting

- Given sufficient data, NNs are well suited to the task of forecasting.

- NNs excel at pattern recognition and forecasting from pattern clusters.

- They allow modelling of complex nonlinear relationships between the response variable and its predictors.

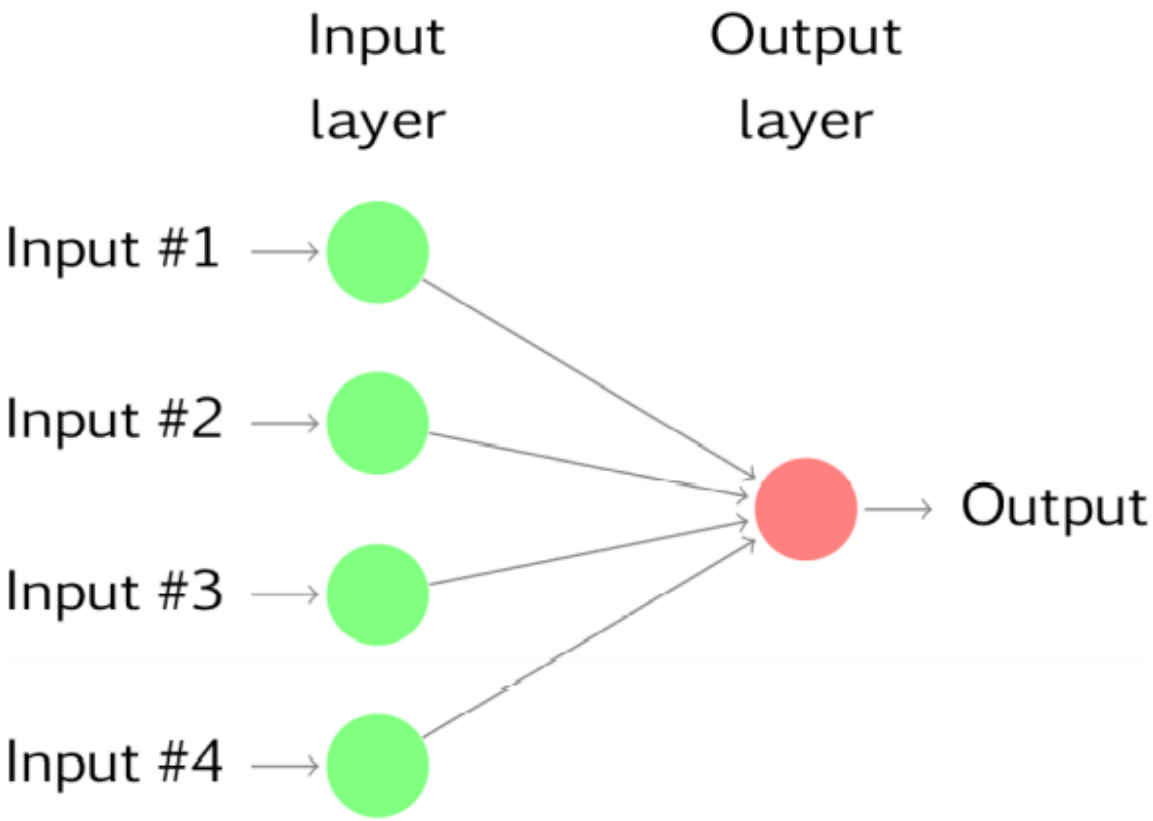
Architecture of NNs

- The general architecture of NNs consists of an interconnected system of neurons arranged in a multi-layered structure, (i.e.) an input layer, an output layer and one or more optional hidden layers.

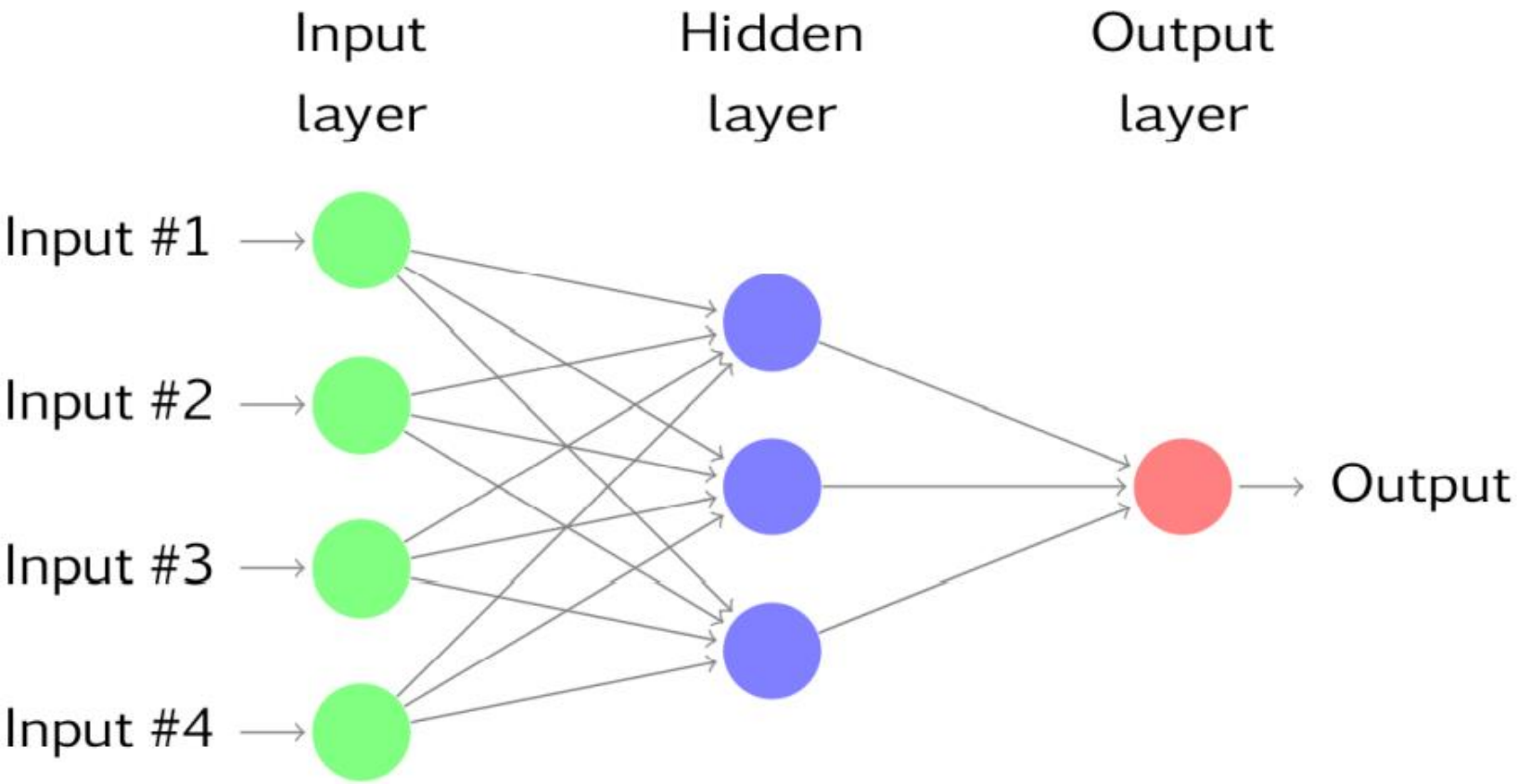
- The simplest networks contain no hidden layers and are equivalent to linear regressions.

- The number of inputs generally correspond to the number of predictor variables in the model.

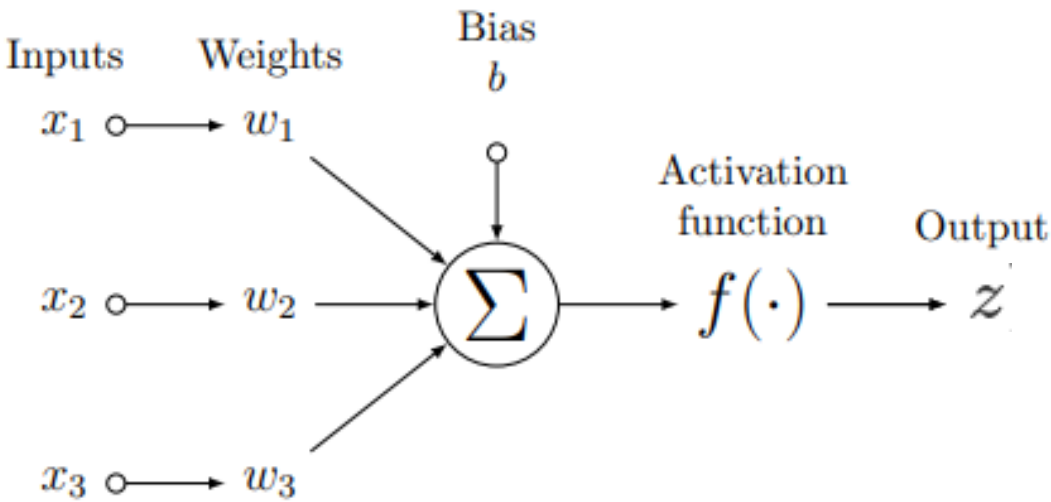
A Simple NN


Input
layer
Output
layer
Input #1
Input #2
Input #3
Input #4
Output

An NN with one Hidden Layer



Structure of a Neuron



Working of Simple NNs

- Each input is initially assigned a weight.
- A linear combination of these weighted inputs along with a bias term is fed into the activation function (sigmoid, logistic, arctan, etc.,), which then transforms it into a scaled output between 0 and 1.
- The deviation of this output from its target value is then used to adjust the weights.
- By using a sufficiently large number of observations, the weights (regression coefficients) corresponding to each input may be optimally tuned.

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Thank You !!!